

# BFS Traversal

Step 1: Start at A (Seeking target node K)

## Legend



Complete



Processing



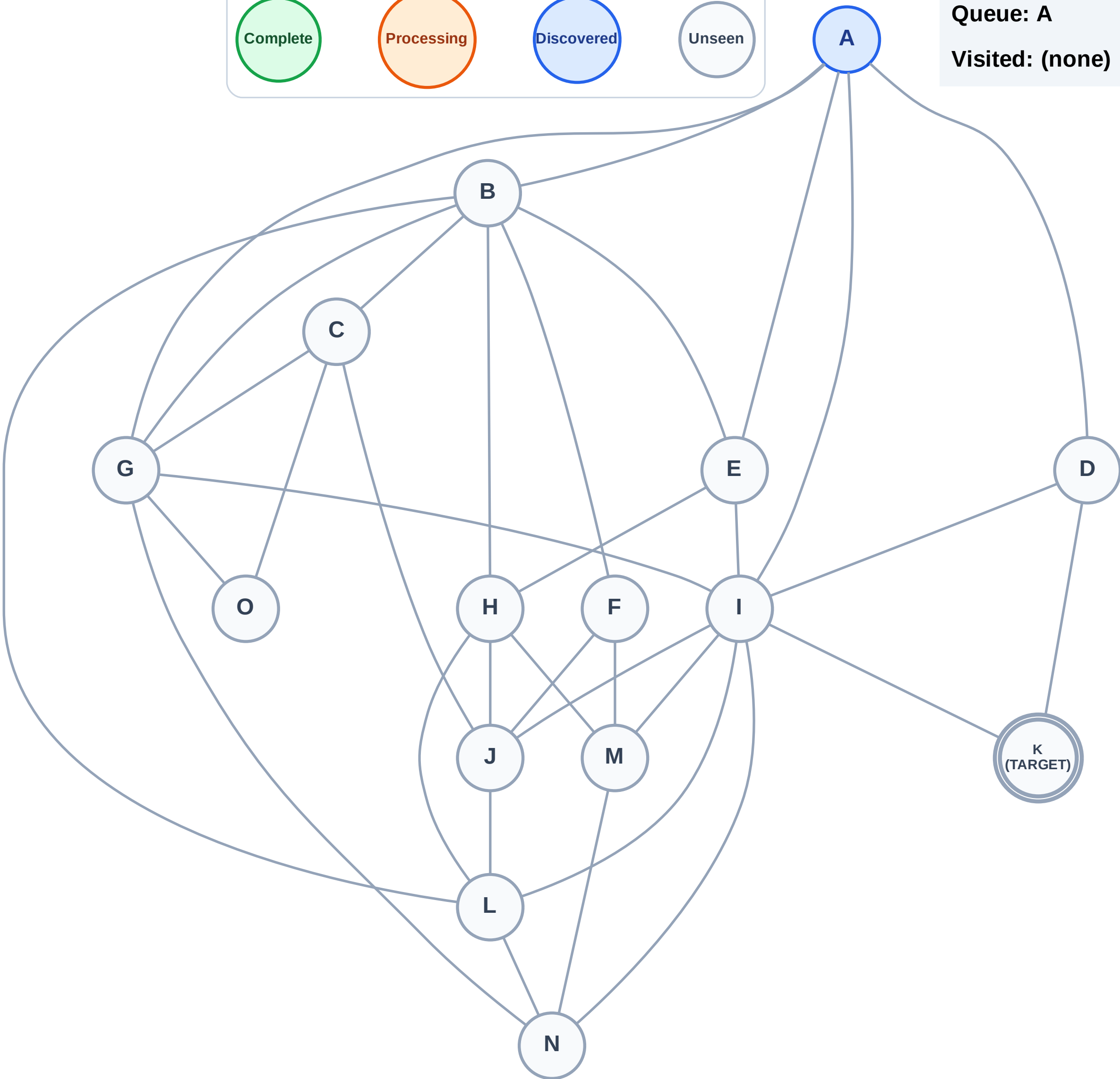
Discovered



Unseen

Queue: A

Visited: (none)



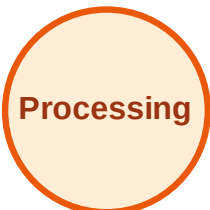
# BFS Traversal

Step 2: Process node A

## Legend



Complete



Processing



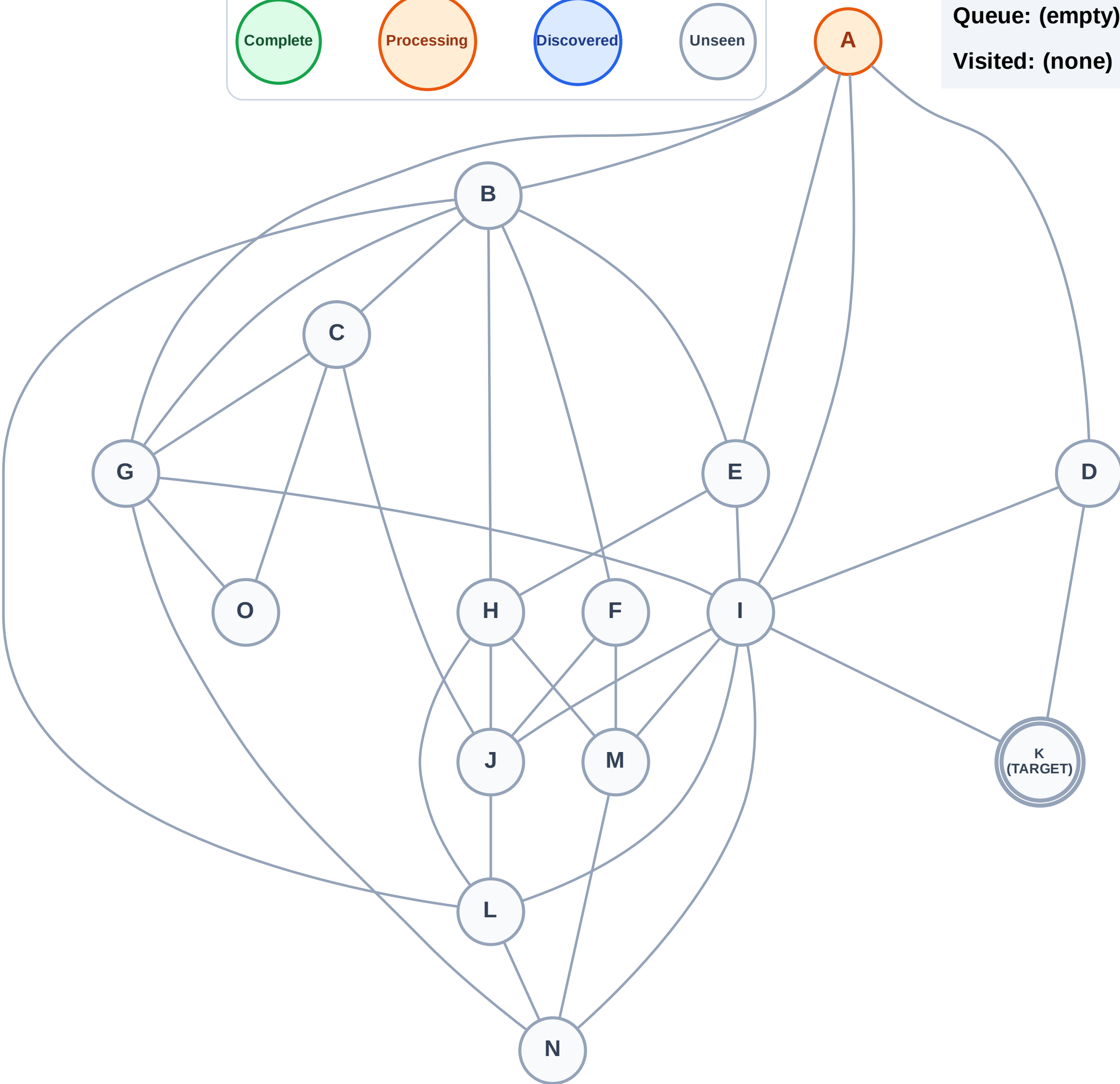
Discovered



Unseen

Queue: (empty)

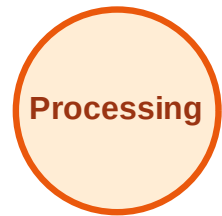
Visited: (none)



# BFS Traversal

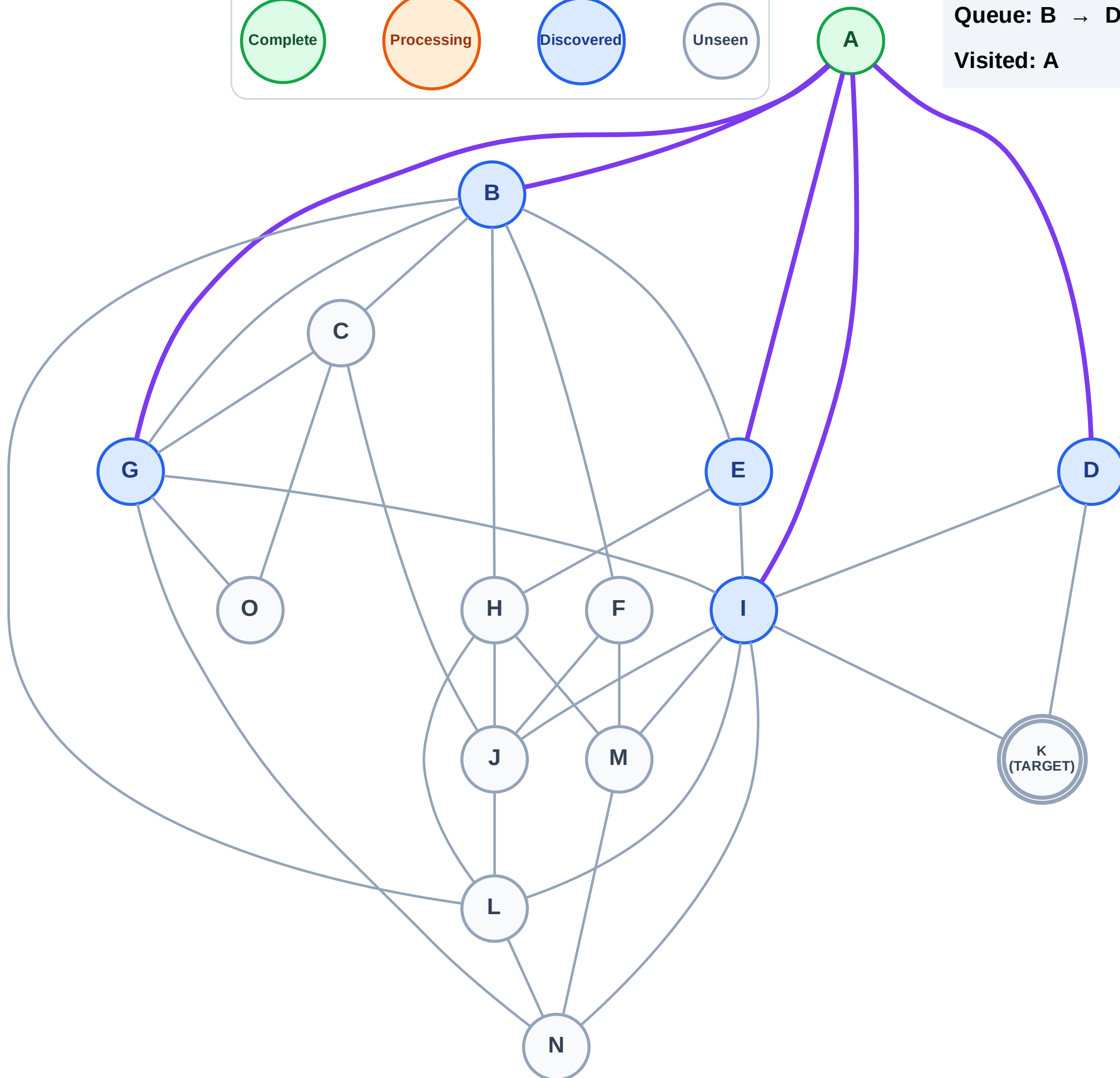
Step 3: Discover B, D, E, G, I from A

## Legend



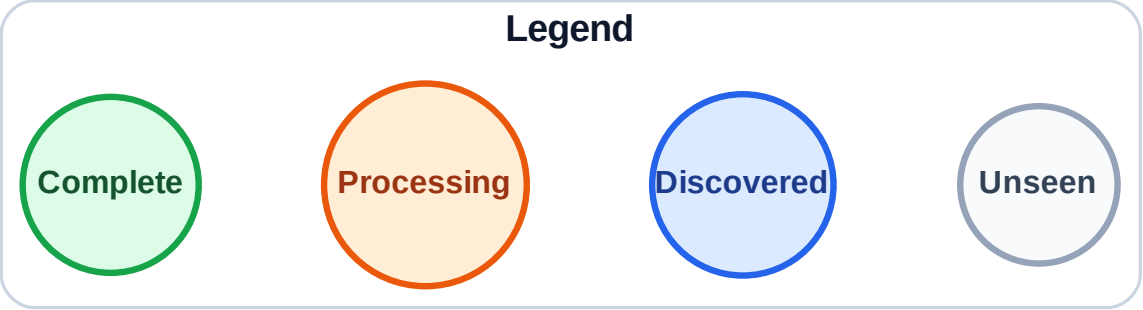
Queue: B → D → E → G → I

Visited: A

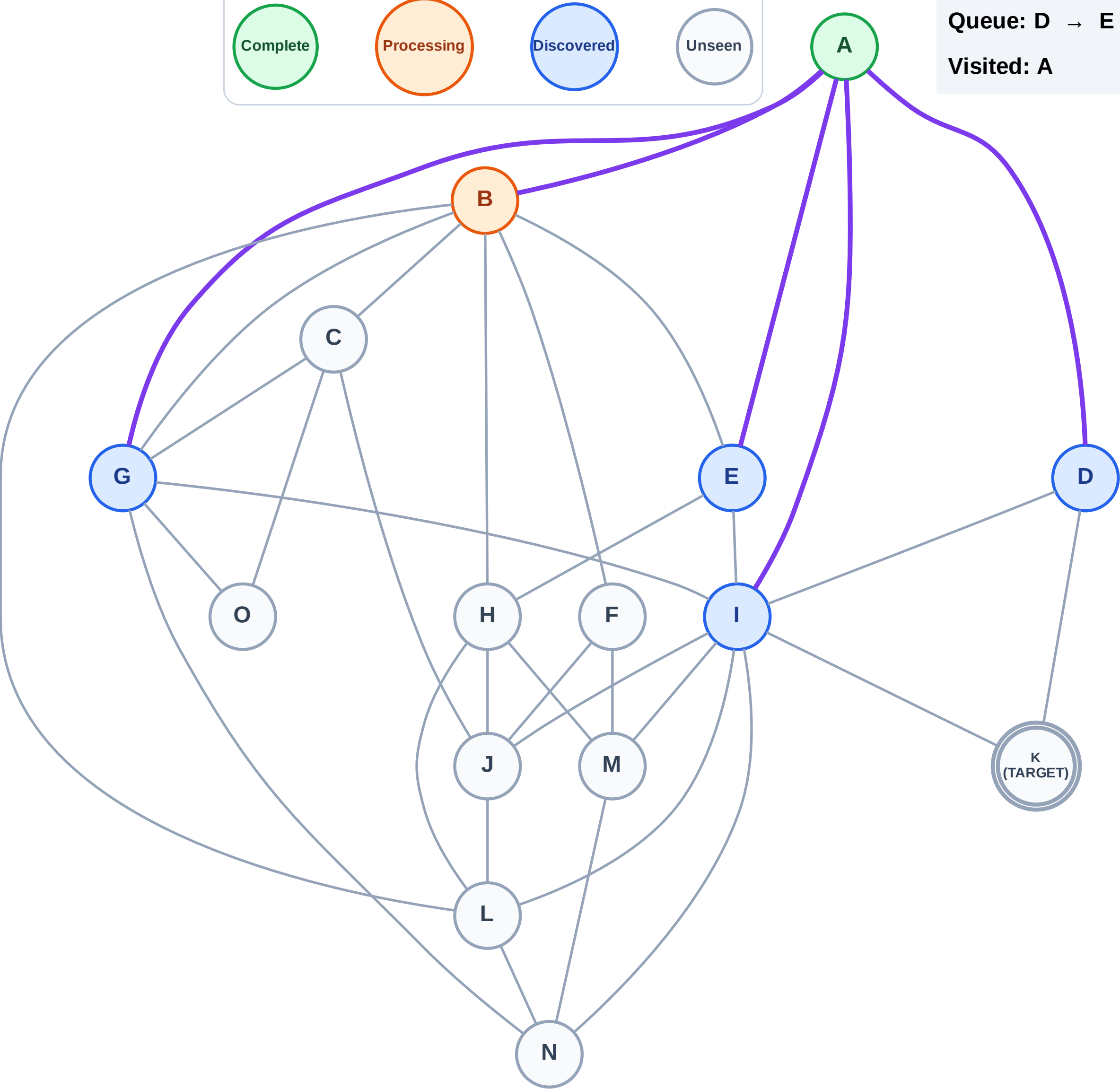


# BFS Traversal

Step 4: Process node B



Queue: D → E → G → I  
Visited: A



# BFS Traversal

Step 5: Discover C, F, H, L from B

## Legend



Complete



Processing



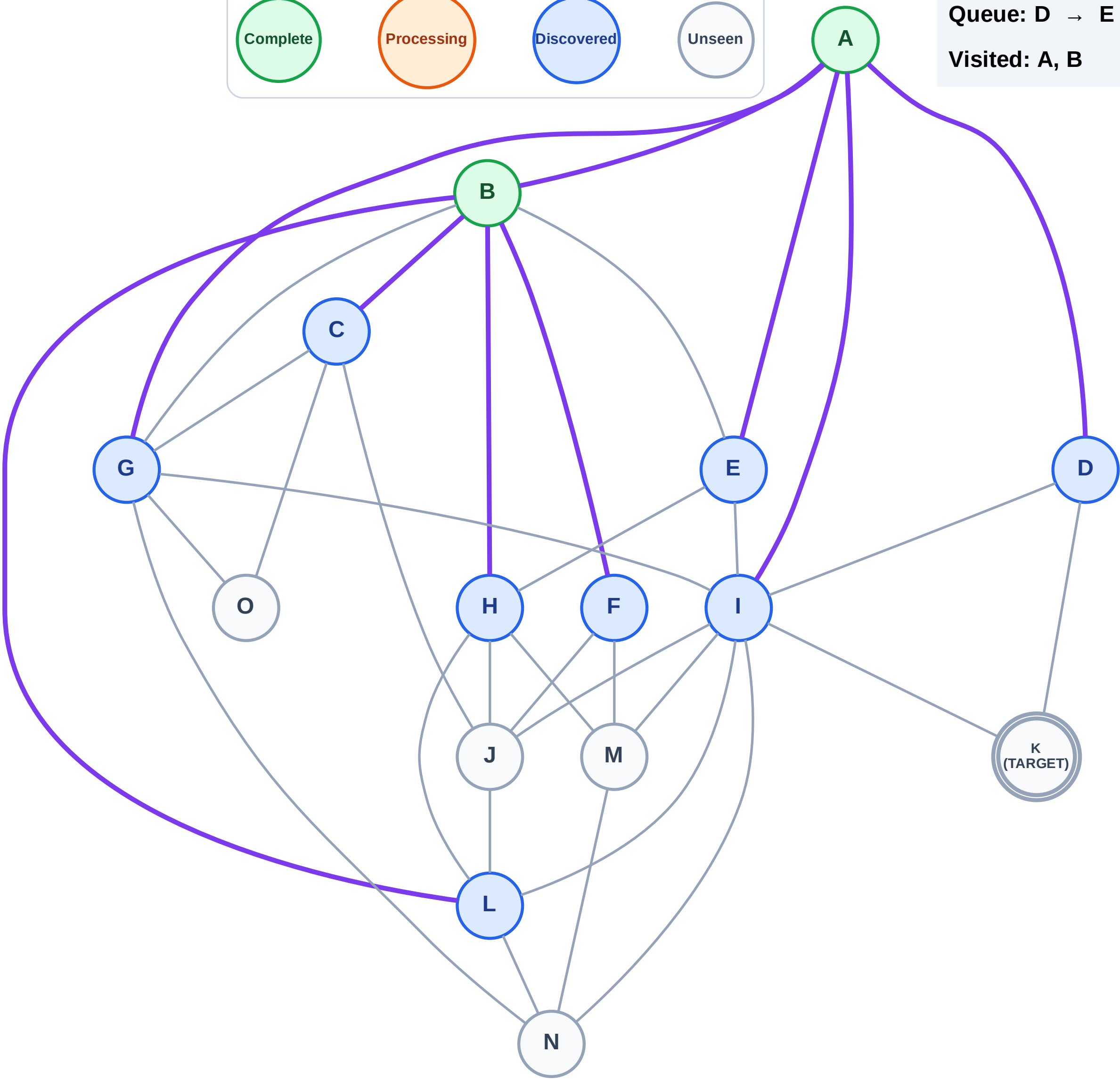
Discovered



Unseen

Queue: D → E → G → I → C → F → H → L

Visited: A, B



# BFS Traversal

Step 6: Process node D

Legend

Complete

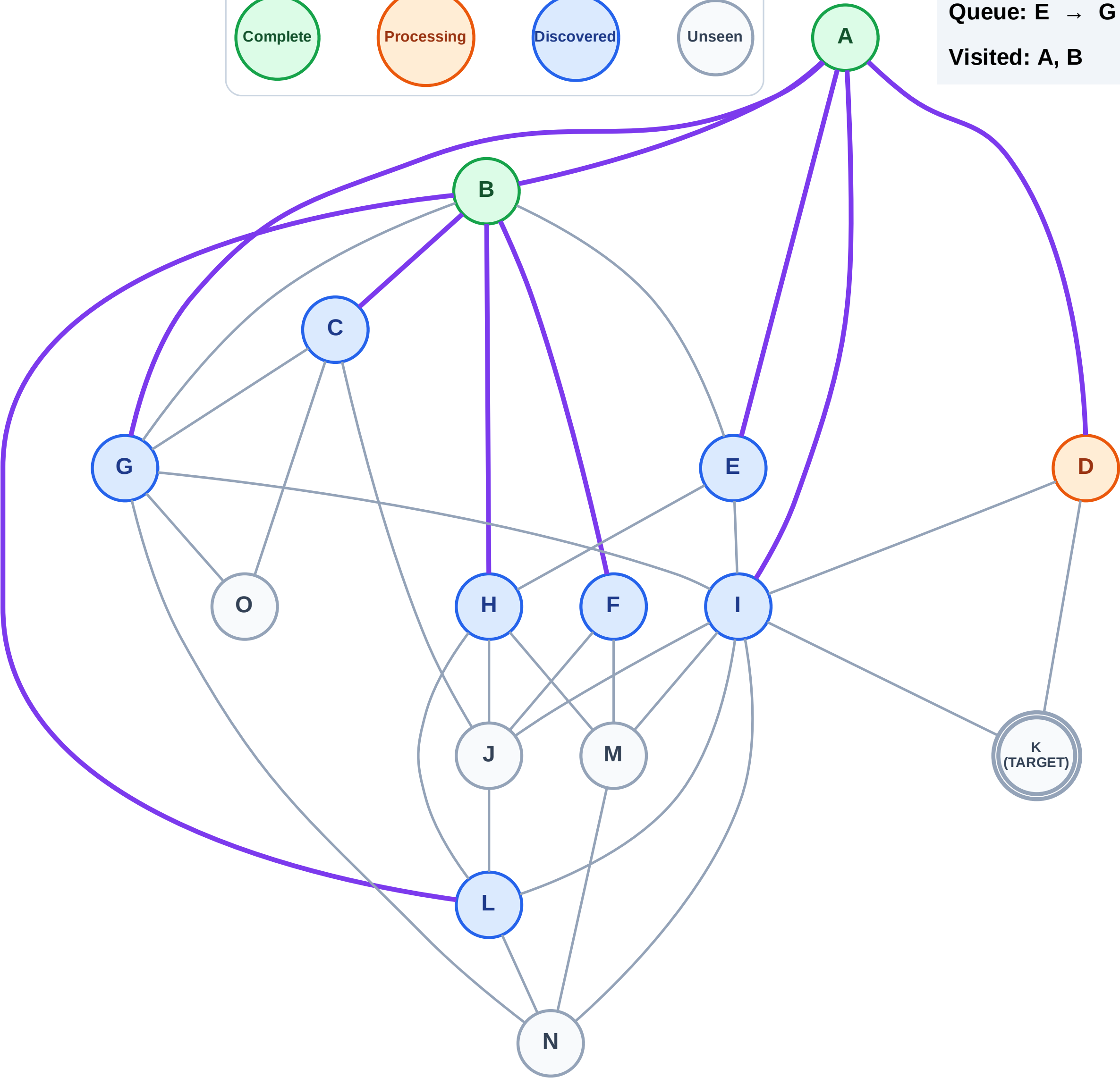
Processing

Discovered

Unseen

Queue: E → G → I → C → F → H → L

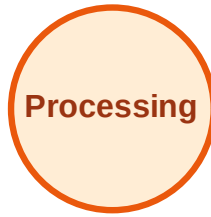
Visited: A, B



# BFS Traversal

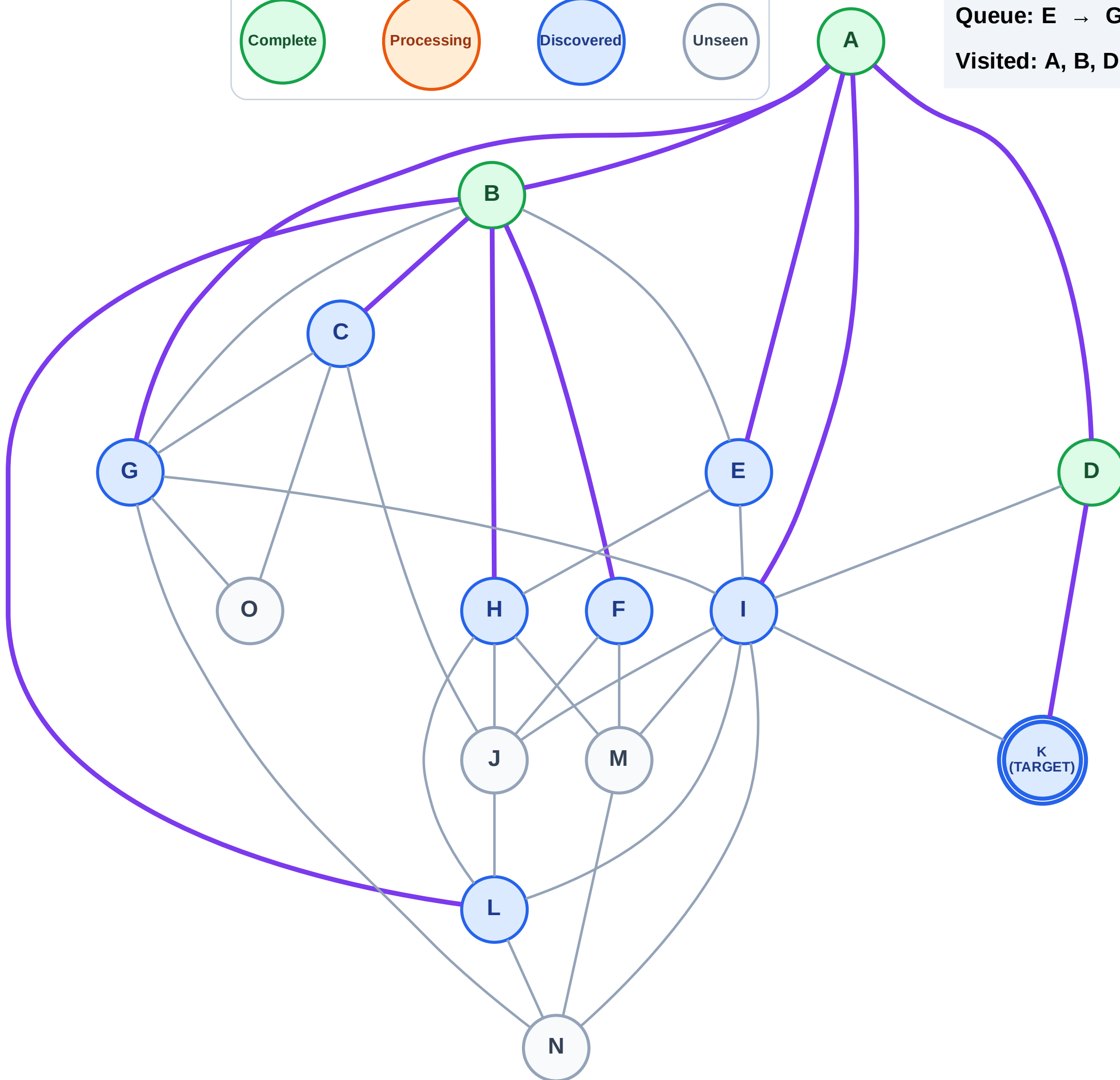
Step 7: Discover K from D

## Legend



Queue: E → G → I → C → F → H → L → K

Visited: A, B, D



# BFS Traversal

Step 8: Process node E

Legend

Complete

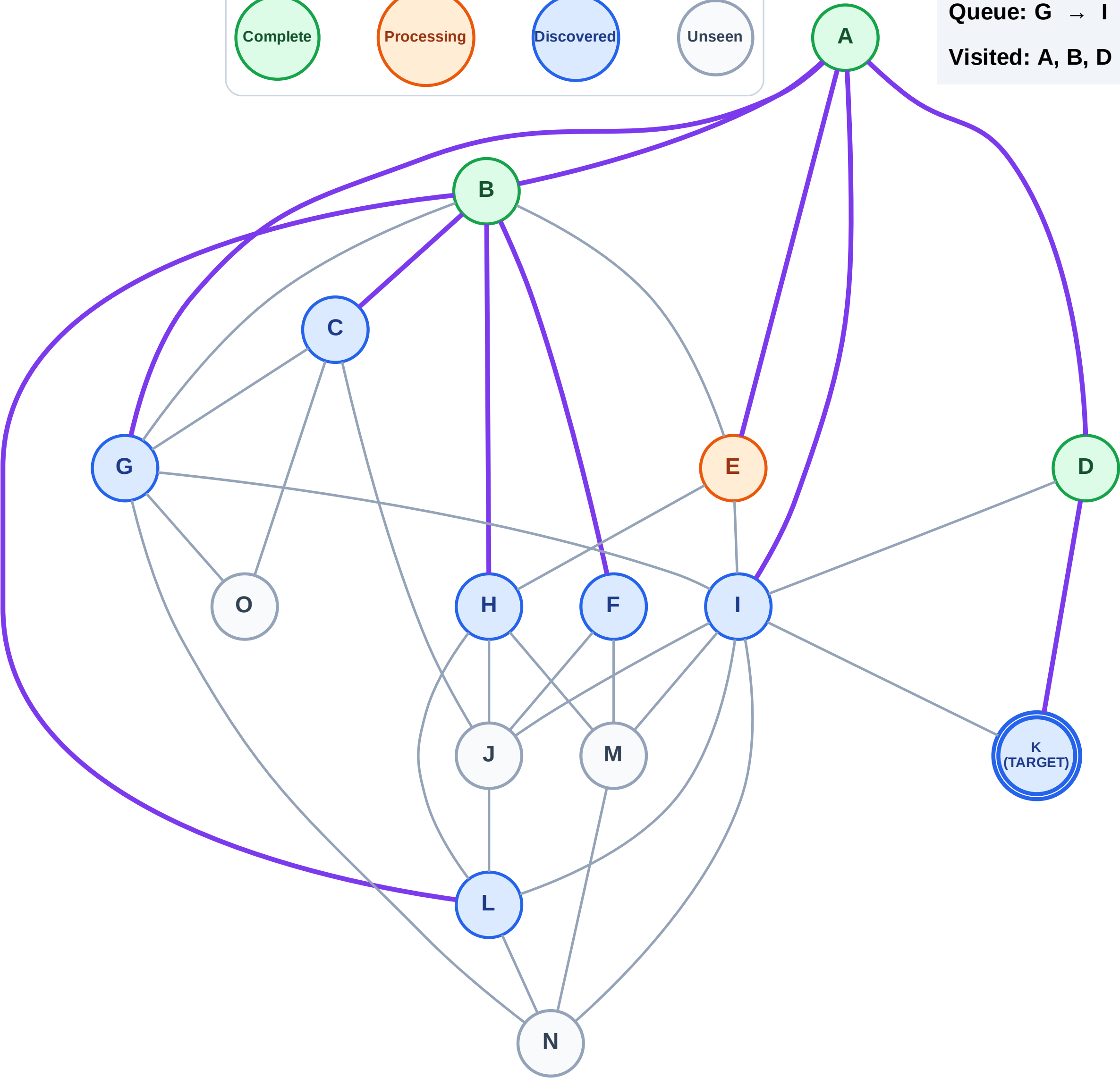
Processing

Discovered

Unseen

Queue: G → I → C → F → H → L → K

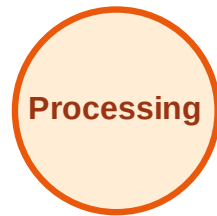
Visited: A, B, D



# BFS Traversal

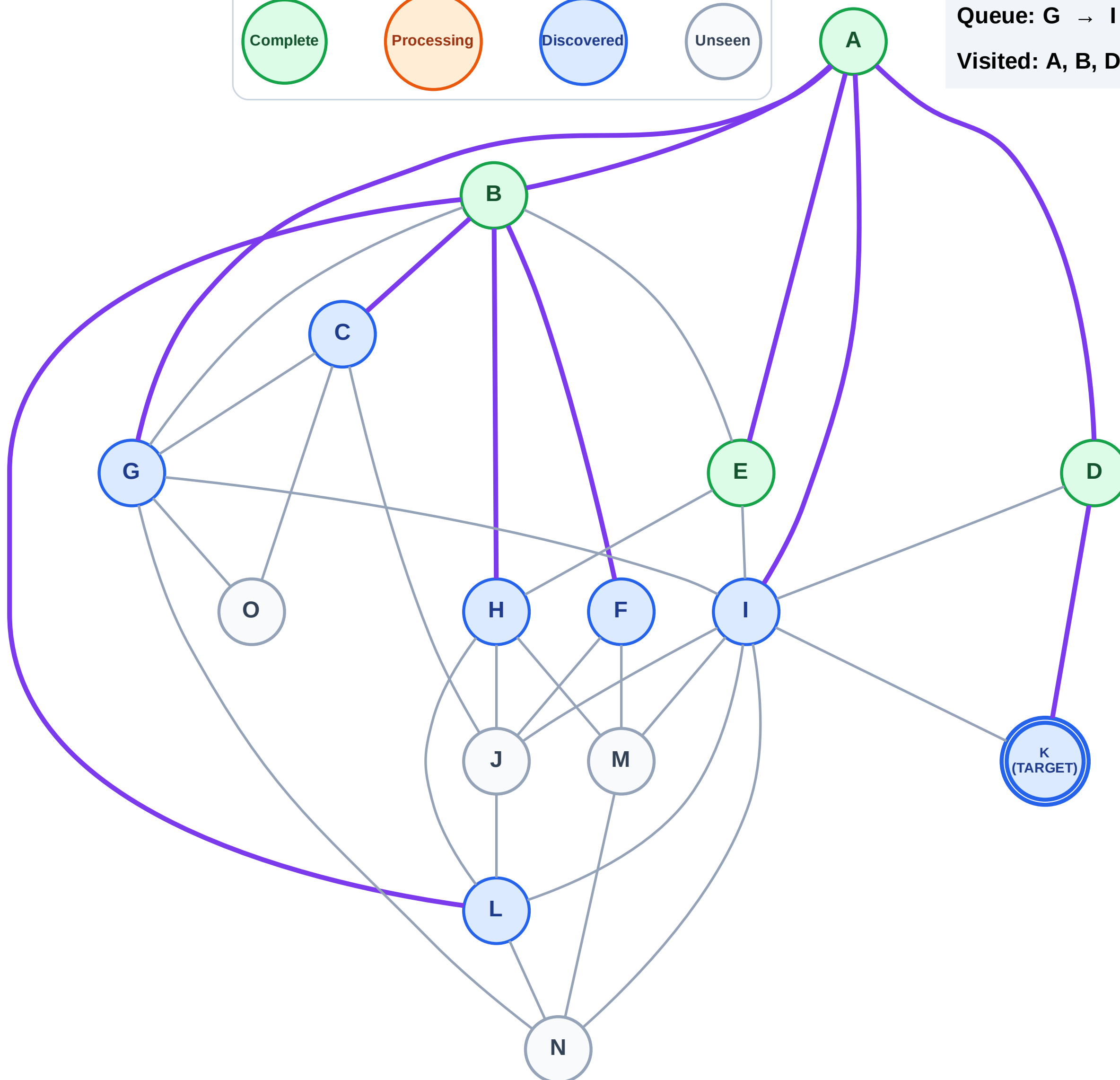
Step 9: No new neighbors from E

## Legend



Queue: G → I → C → F → H → L → K

Visited: A, B, D, E



# BFS Traversal

Step 10: Process node G

## Legend



Complete



Processing



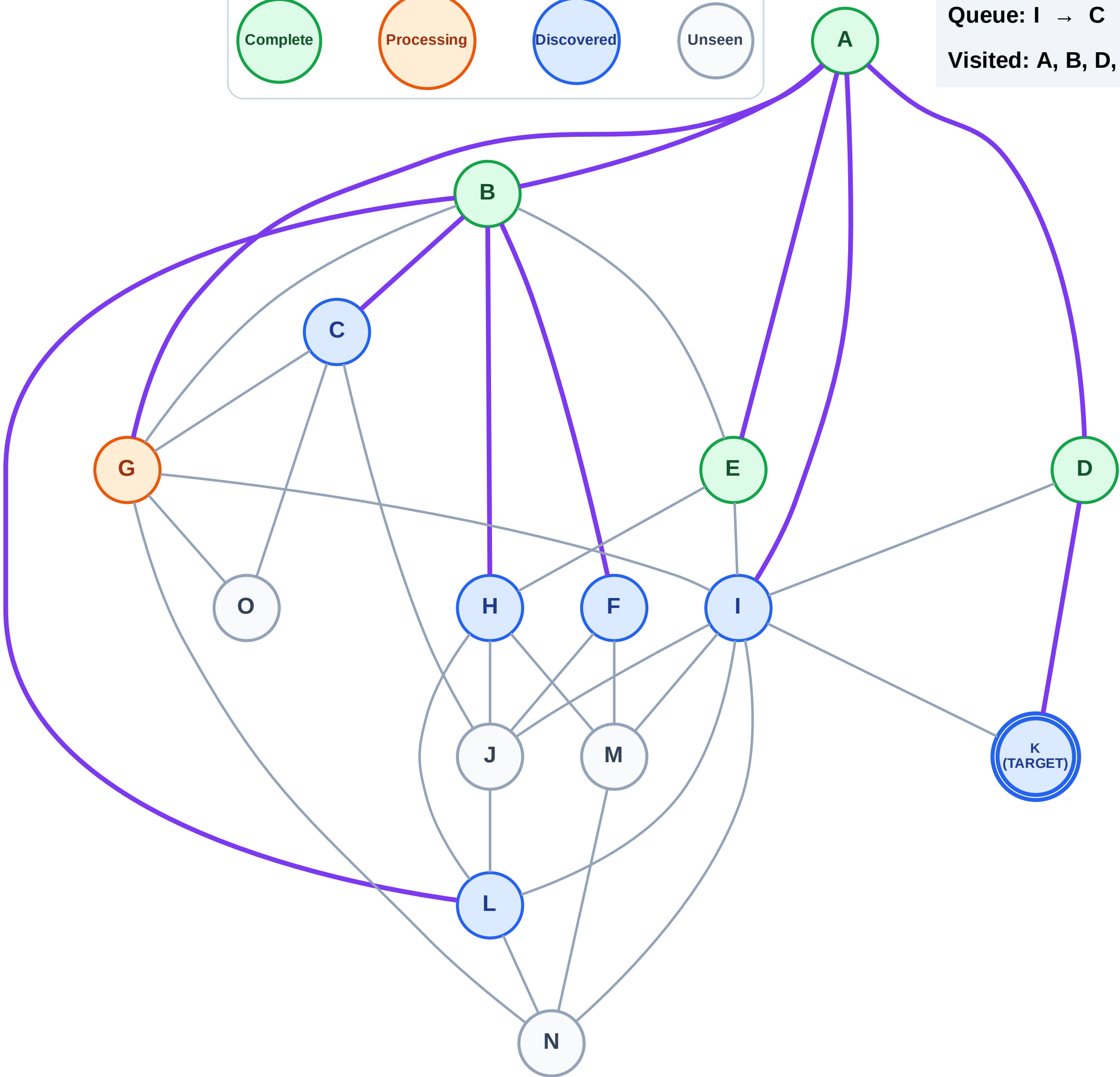
Discovered



Unseen

Queue: I → C → F → H → L → K

Visited: A, B, D, E



# BFS Traversal

Step 11: Discover N, O from G

## Legend



Complete



Processing



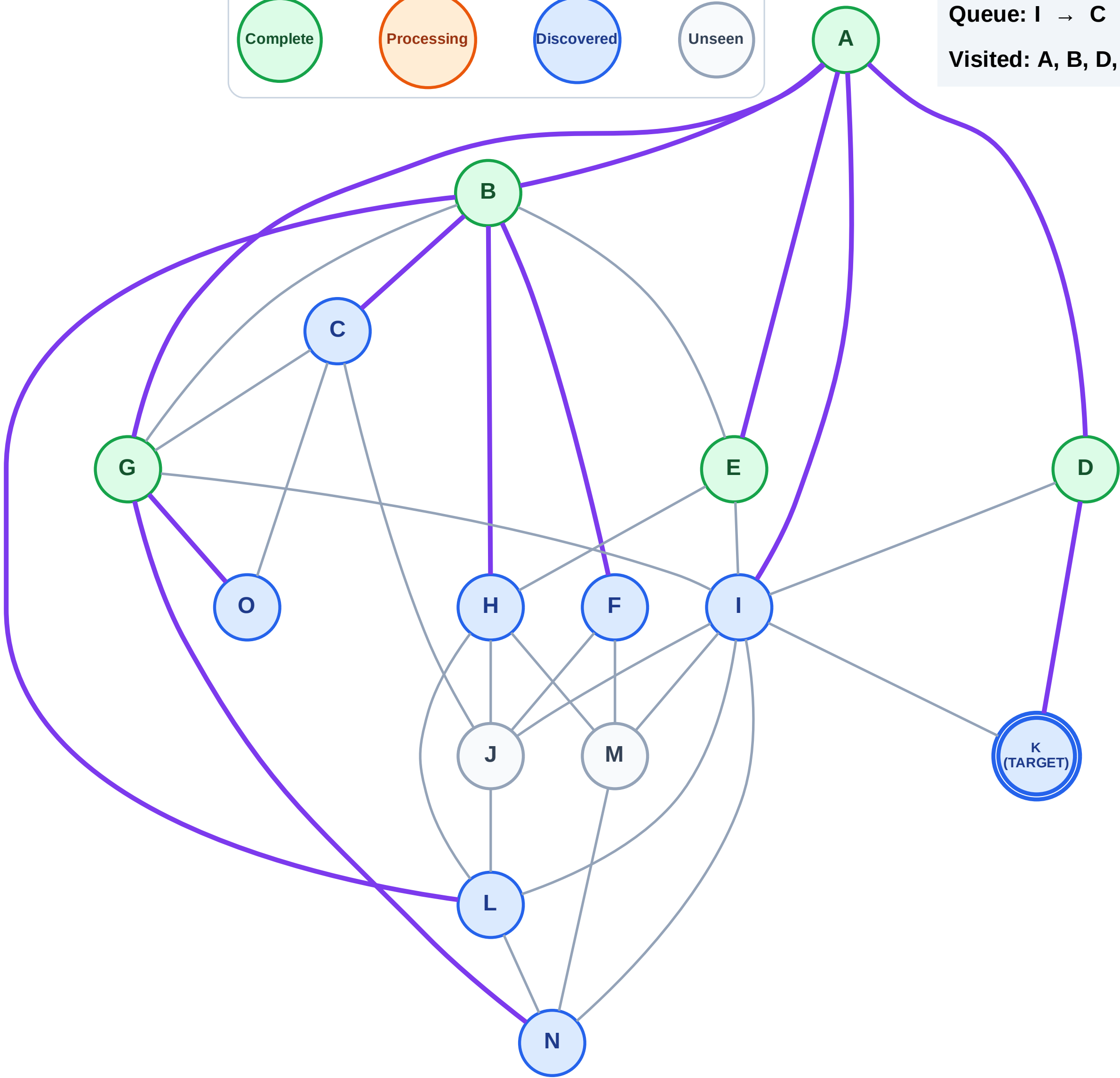
Discovered



Unseen

Queue: I → C → F → H → L → K → N → O

Visited: A, B, D, E, G



# BFS Traversal

Step 12: Process node I

Legend

Complete

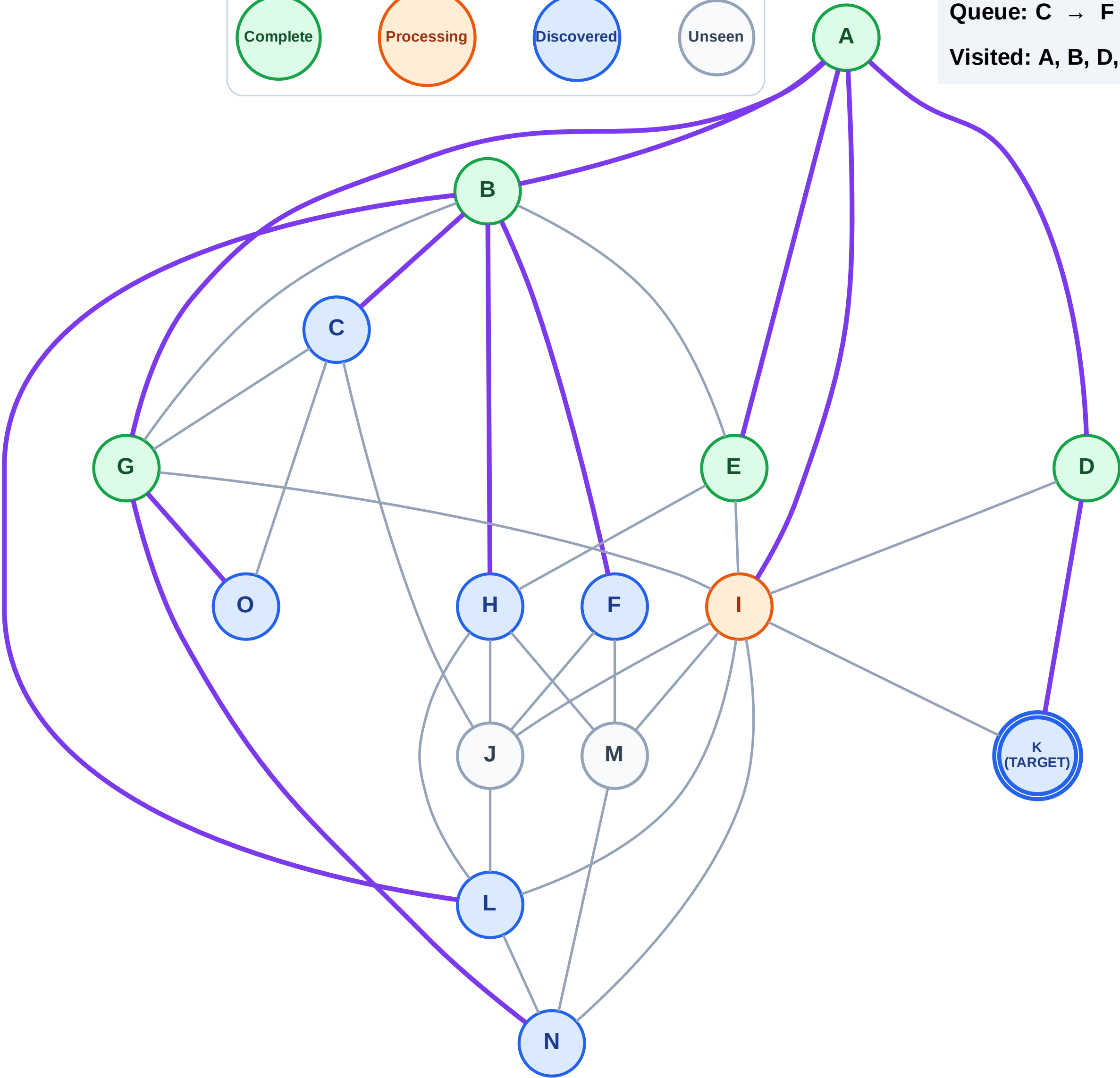
Processing

Discovered

Unseen

Queue: C → F → H → L → K → N → O

Visited: A, B, D, E, G



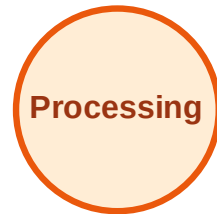
# BFS Traversal

Step 13: Discover J, M from I

## Legend



Complete



Processing



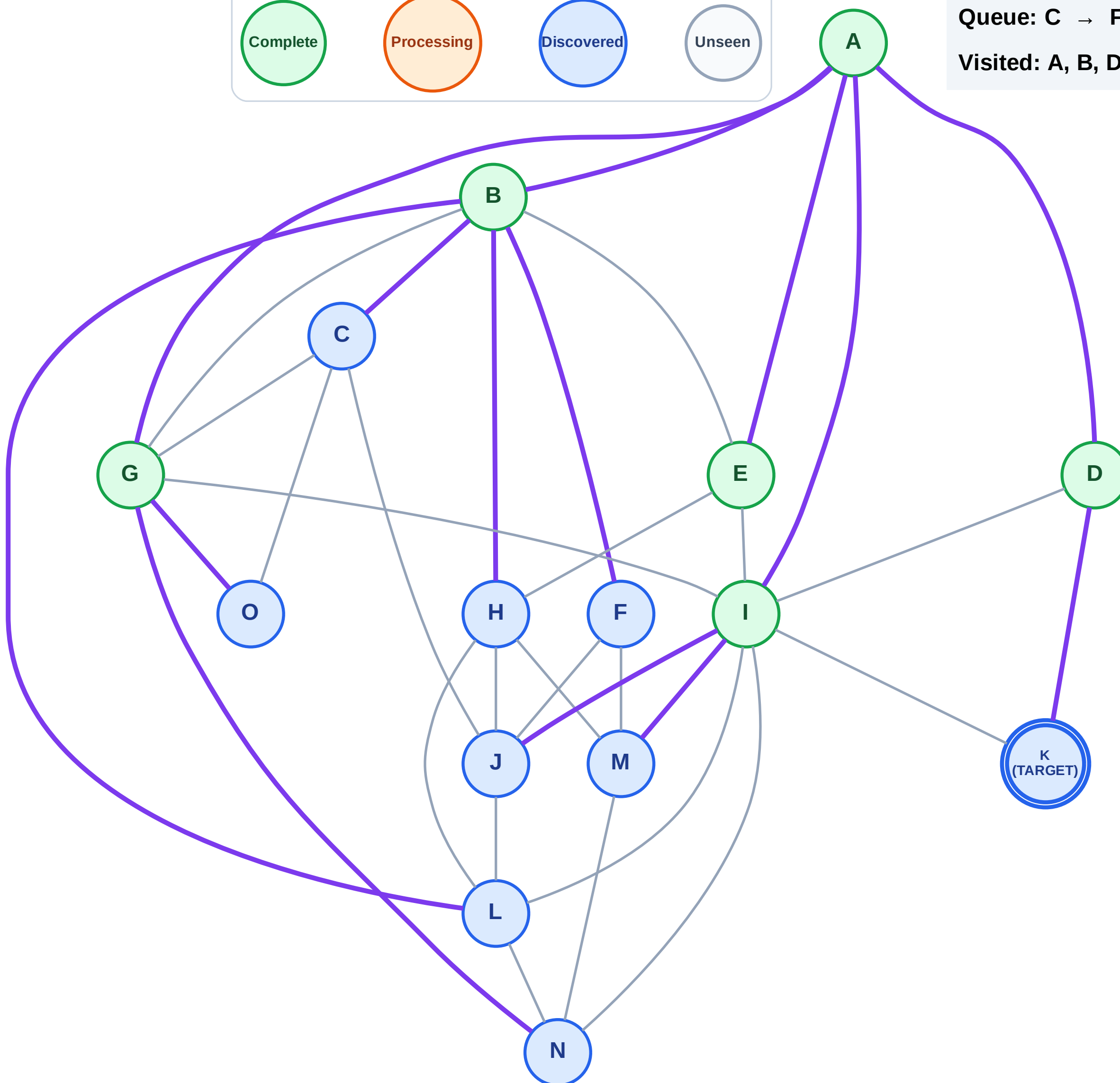
Discovered



Unseen

Queue: C → F → H → L → K → N → O → J → M

Visited: A, B, D, E, G, I



# BFS Traversal

Step 14: Process node C

Legend

Complete

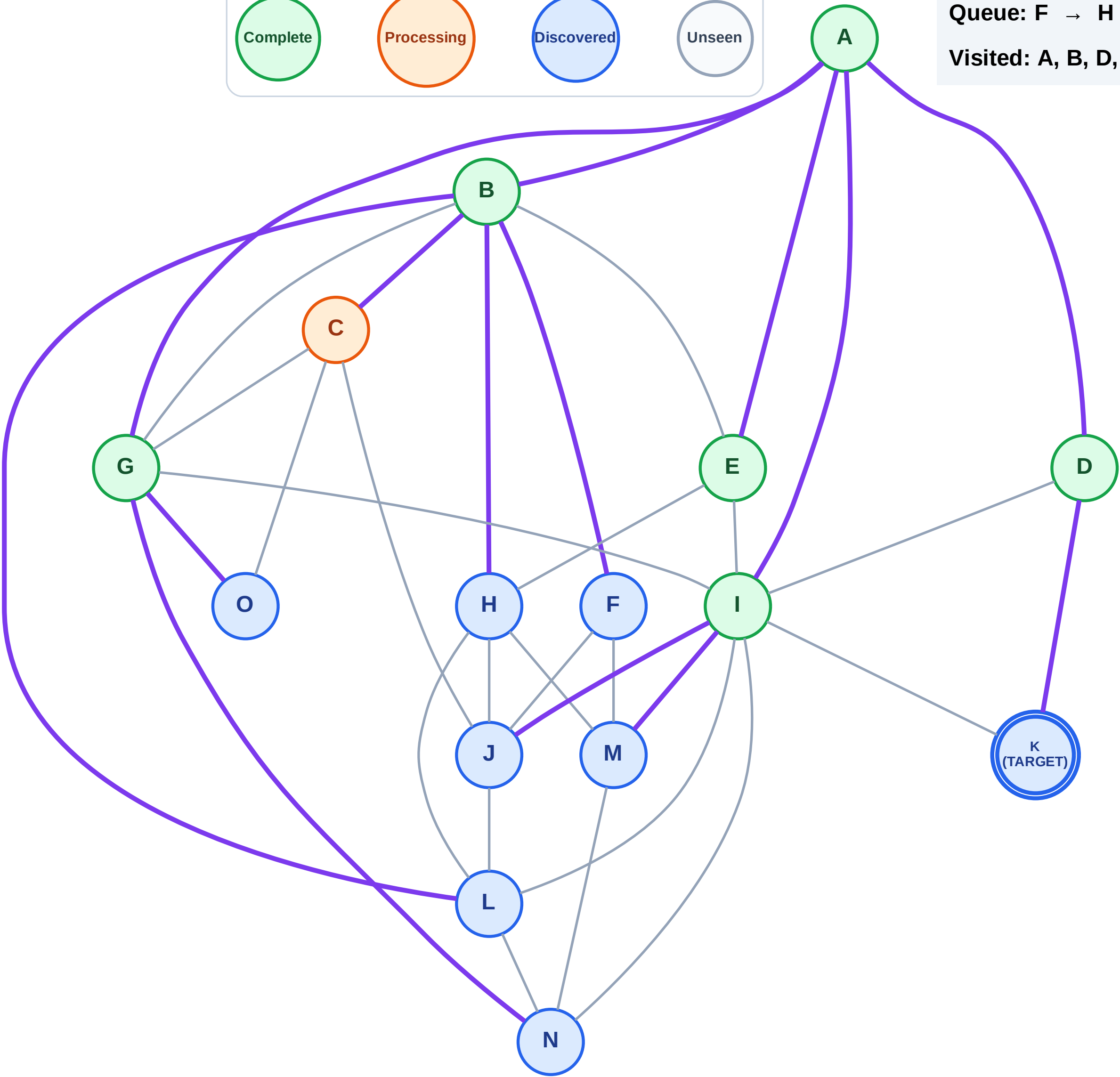
Processing

Discovered

Unseen

Queue: F → H → L → K → N → O → J → M

Visited: A, B, D, E, G, I



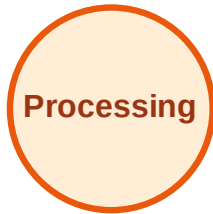
# BFS Traversal

Step 15: No new neighbors from C

## Legend



Complete



Processing



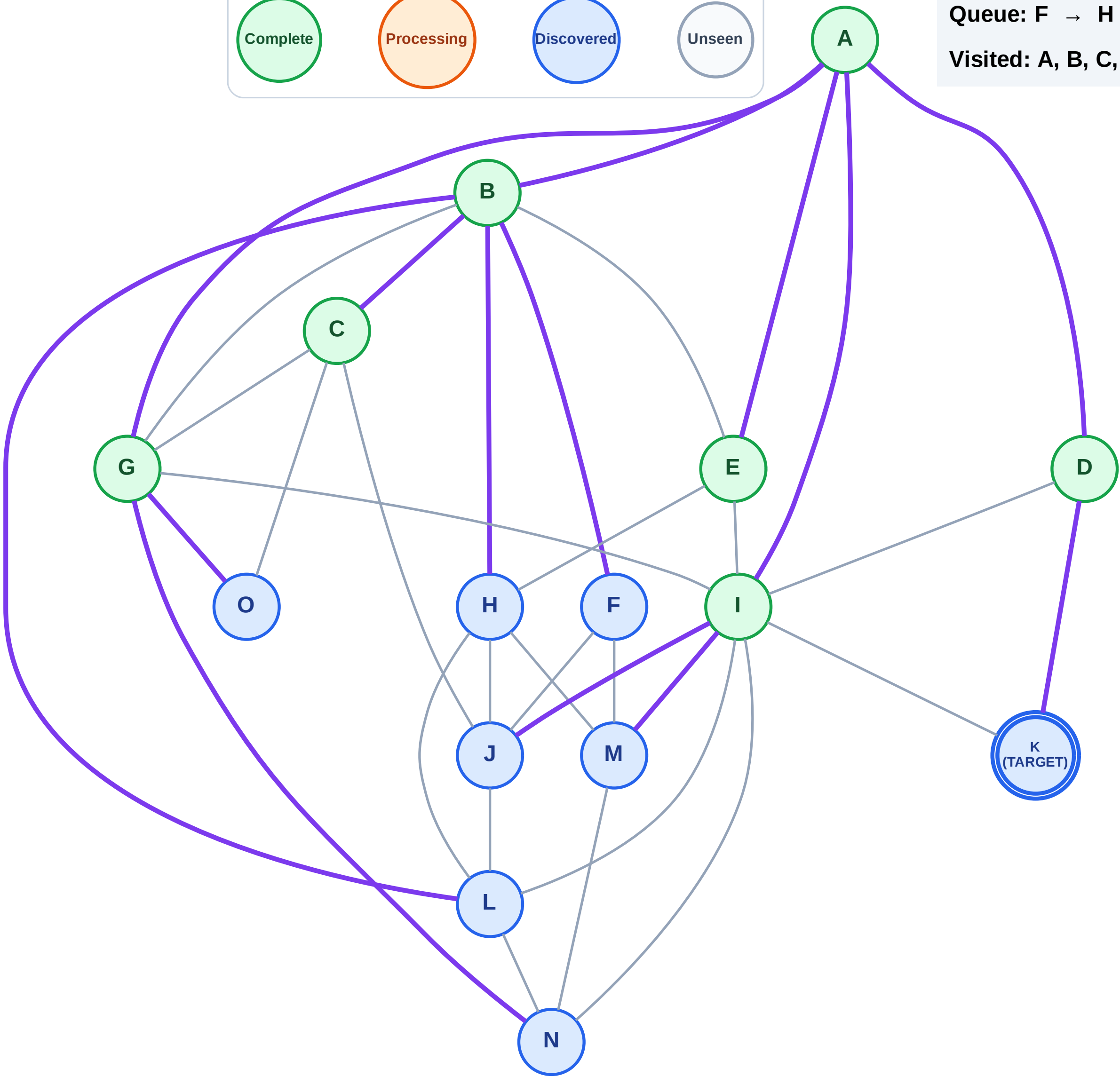
Discovered



Unseen

Queue: F → H → L → K → N → O → J → M

Visited: A, B, C, D, E, G, I



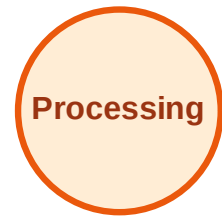
# BFS Traversal

Step 16: Process node F

## Legend



Complete



Processing



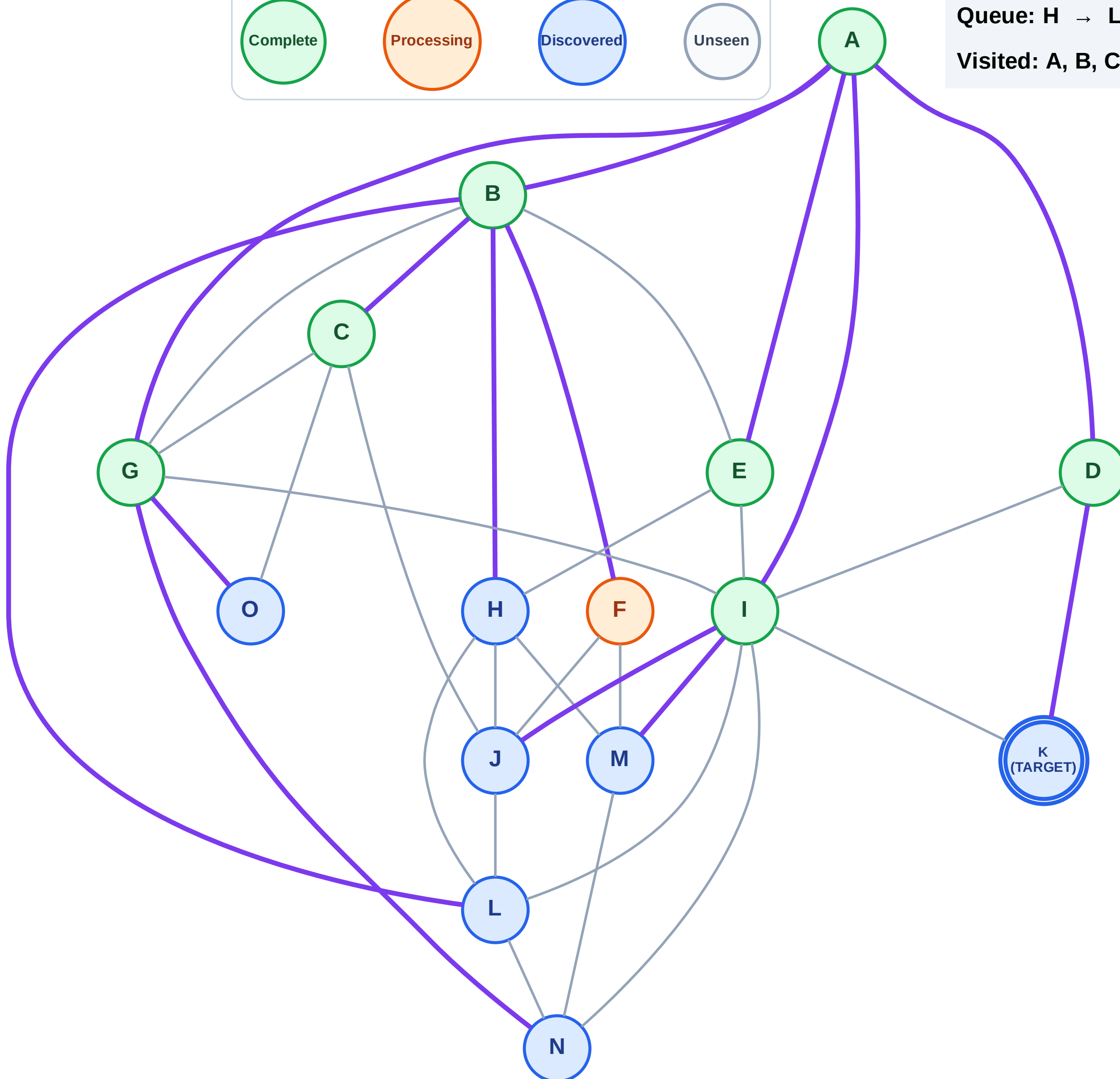
Discovered



Unseen

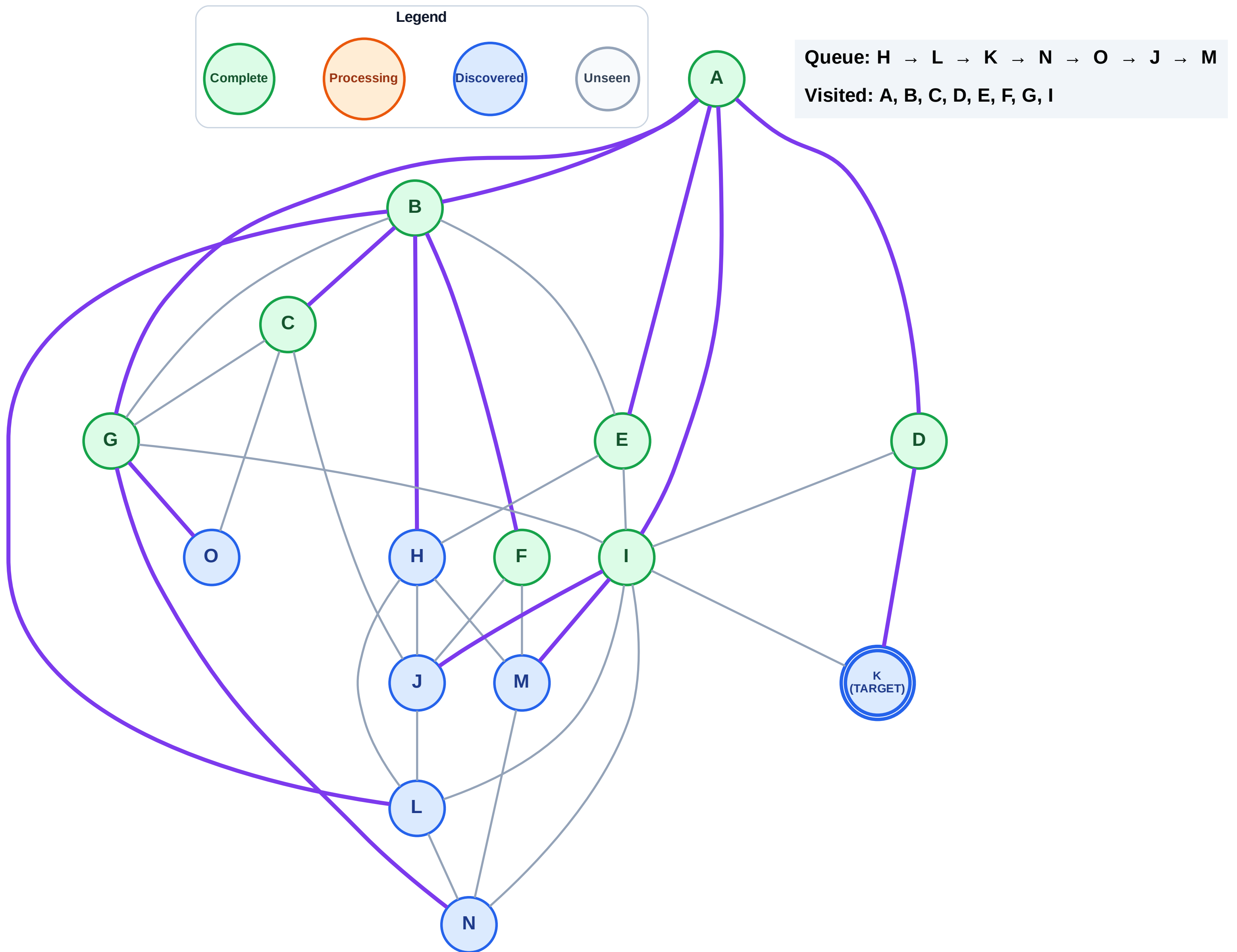
Queue: H → L → K → N → O → J → M

Visited: A, B, C, D, E, G, I



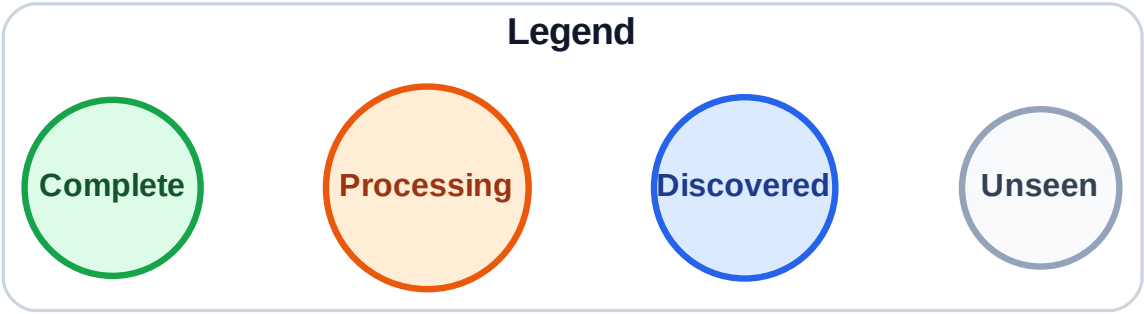
Step 17: No new neighbors from F

Step 17: No new neighbors from F



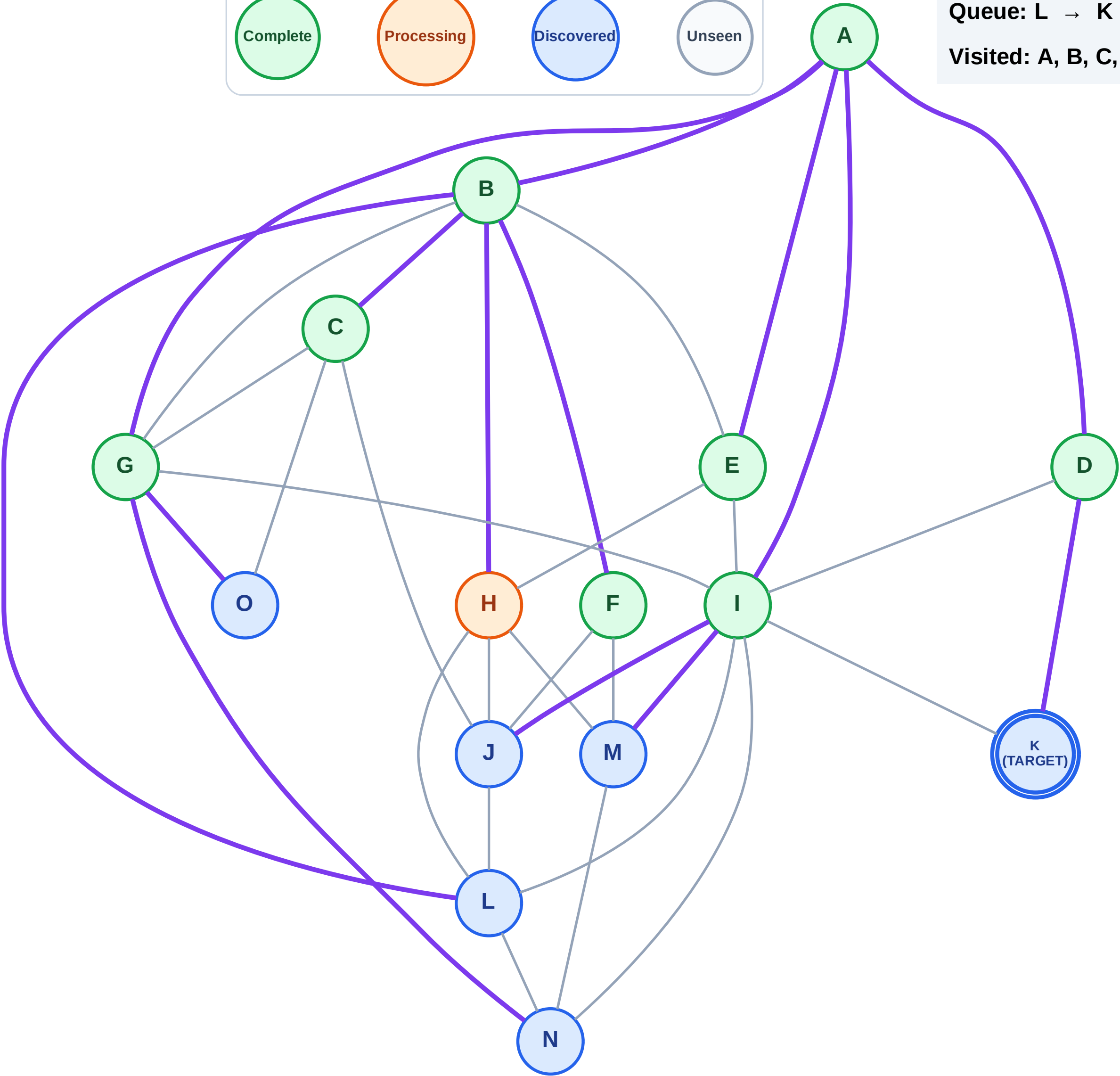
# BFS Traversal

Step 18: Process node H



Queue: L → K → N → O → J → M

Visited: A, B, C, D, E, F, G, I



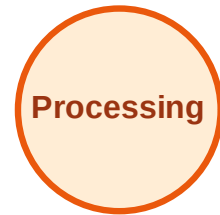
# BFS Traversal

Step 19: No new neighbors from H

## Legend



Complete



Processing



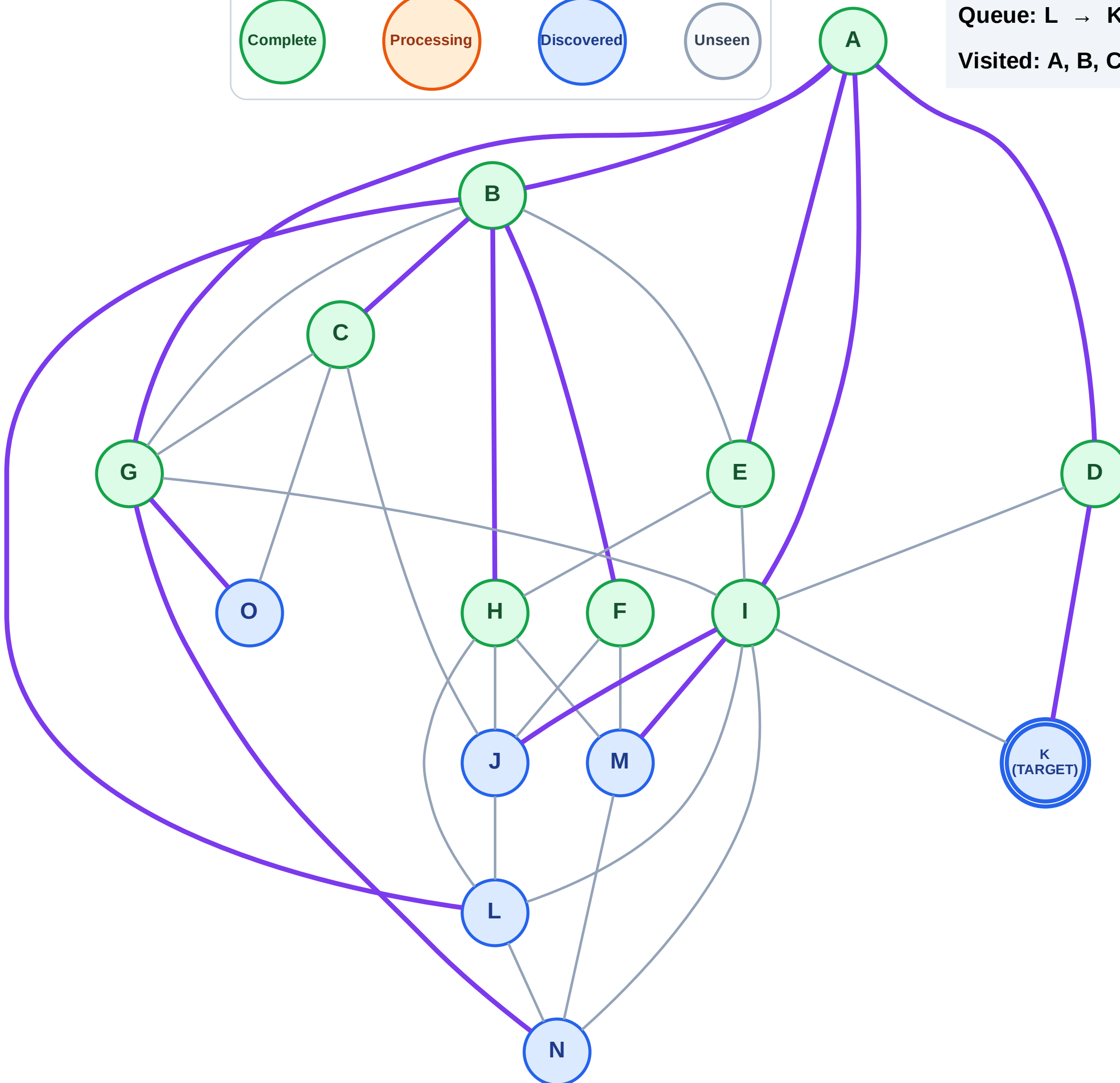
Discovered



Unseen

Queue: L → K → N → O → J → M

Visited: A, B, C, D, E, F, G, H, I



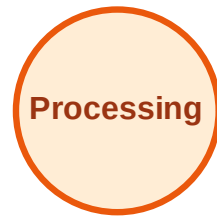
# BFS Traversal

Step 20: Process node L

## Legend



Complete



Processing



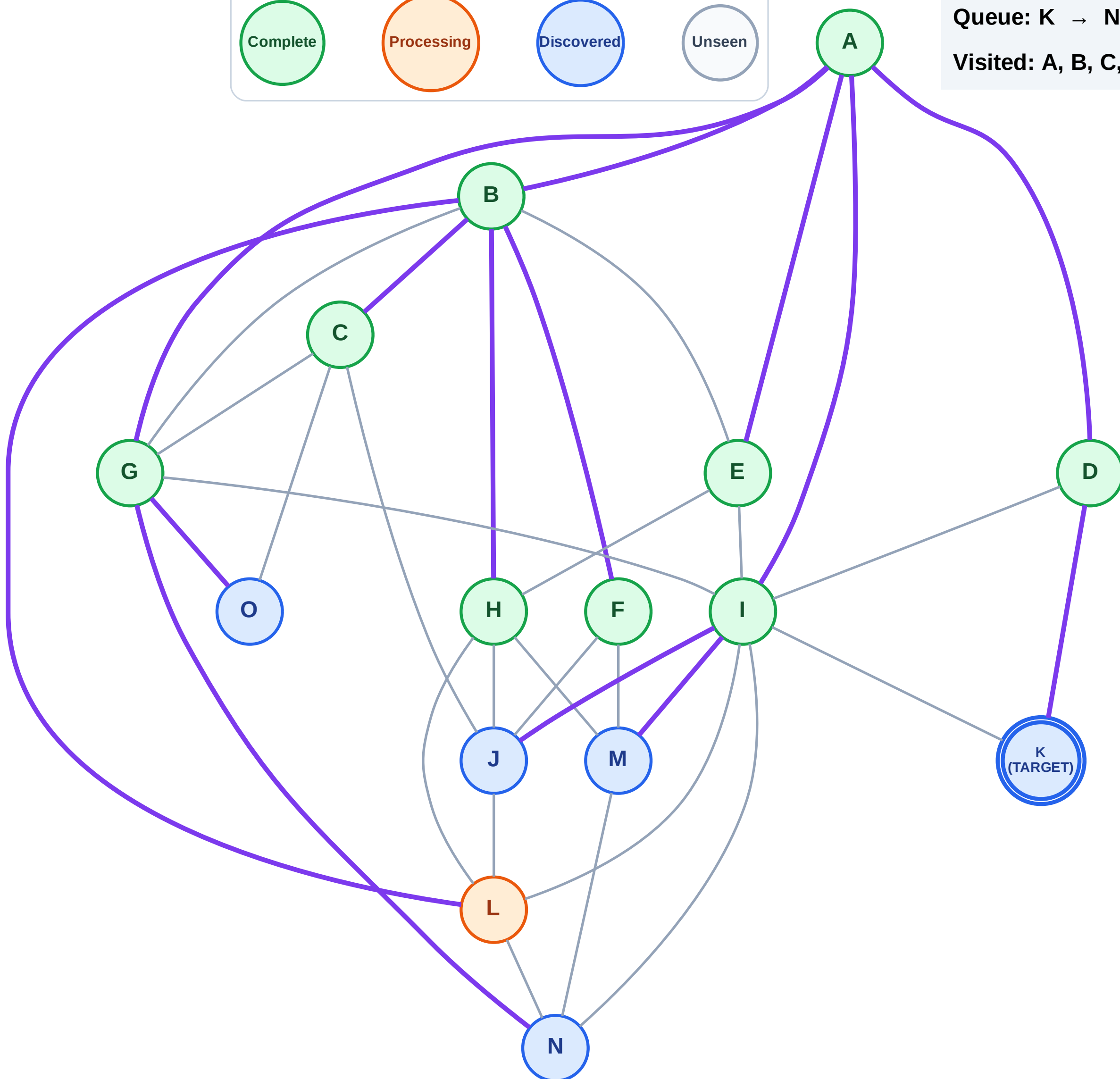
Discovered



Unseen

Queue: K → N → O → J → M

Visited: A, B, C, D, E, F, G, H, I



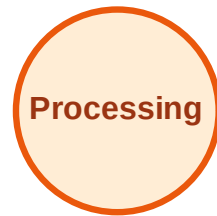
# BFS Traversal

Step 21: No new neighbors from L

## Legend



Complete



Processing



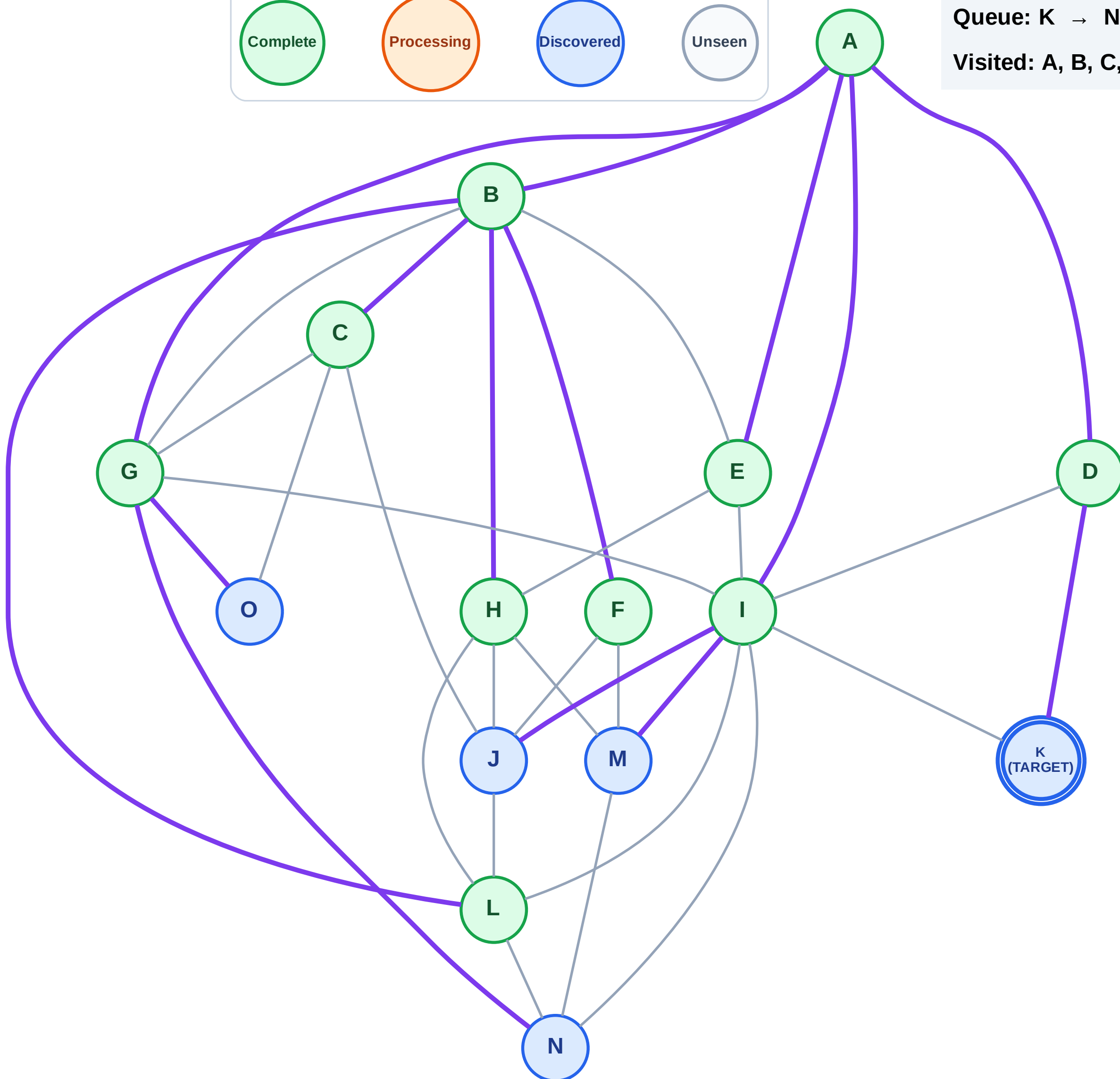
Discovered



Unseen

Queue: K → N → O → J → M

Visited: A, B, C, D, E, F, G, H, I, L



# BFS Traversal

Step 22: Process node K

## Legend



Complete



Processing



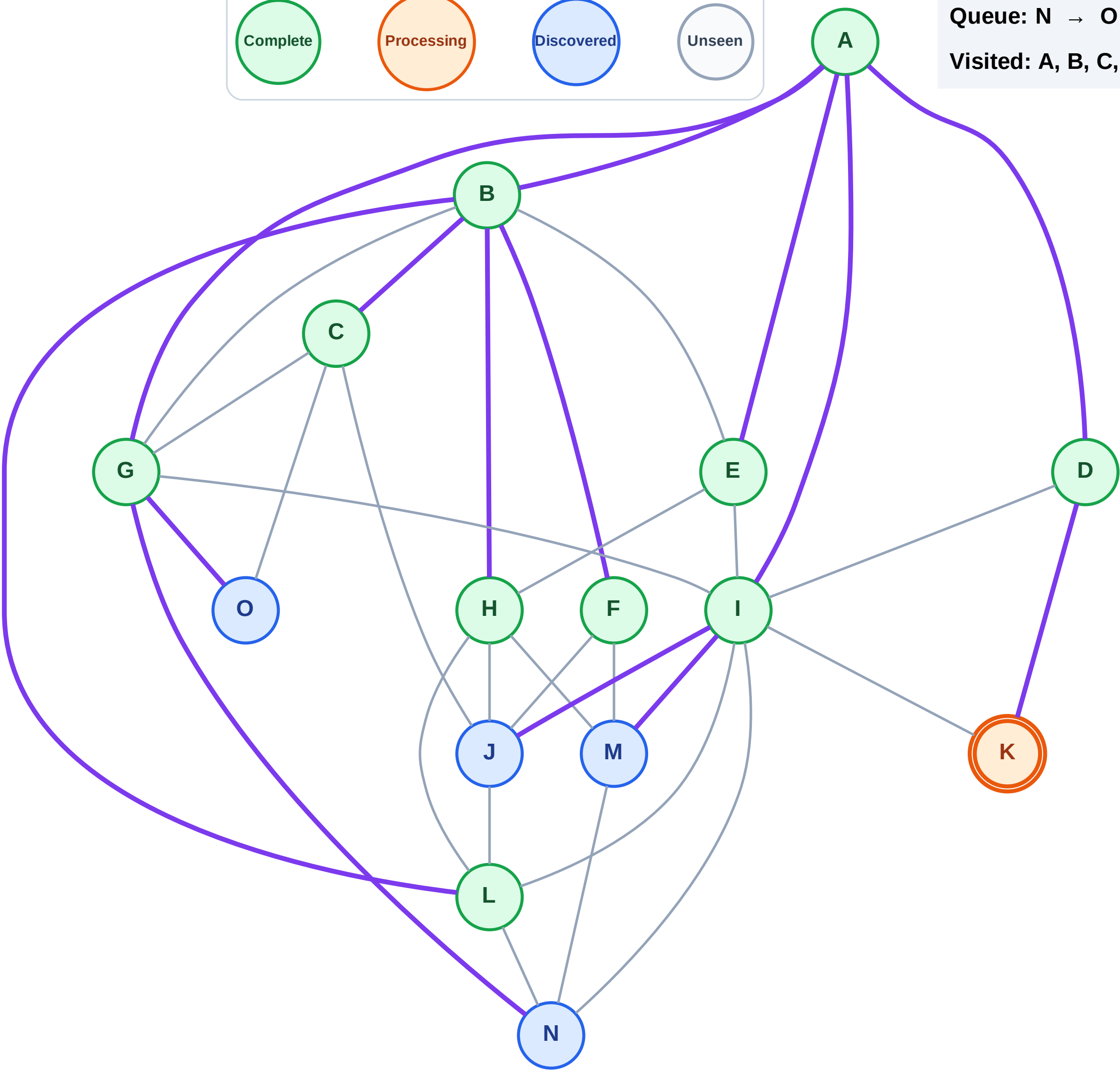
Discovered



Unseen

Queue: N → O → J → M

Visited: A, B, C, D, E, F, G, H, I, L



# BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend

Complete

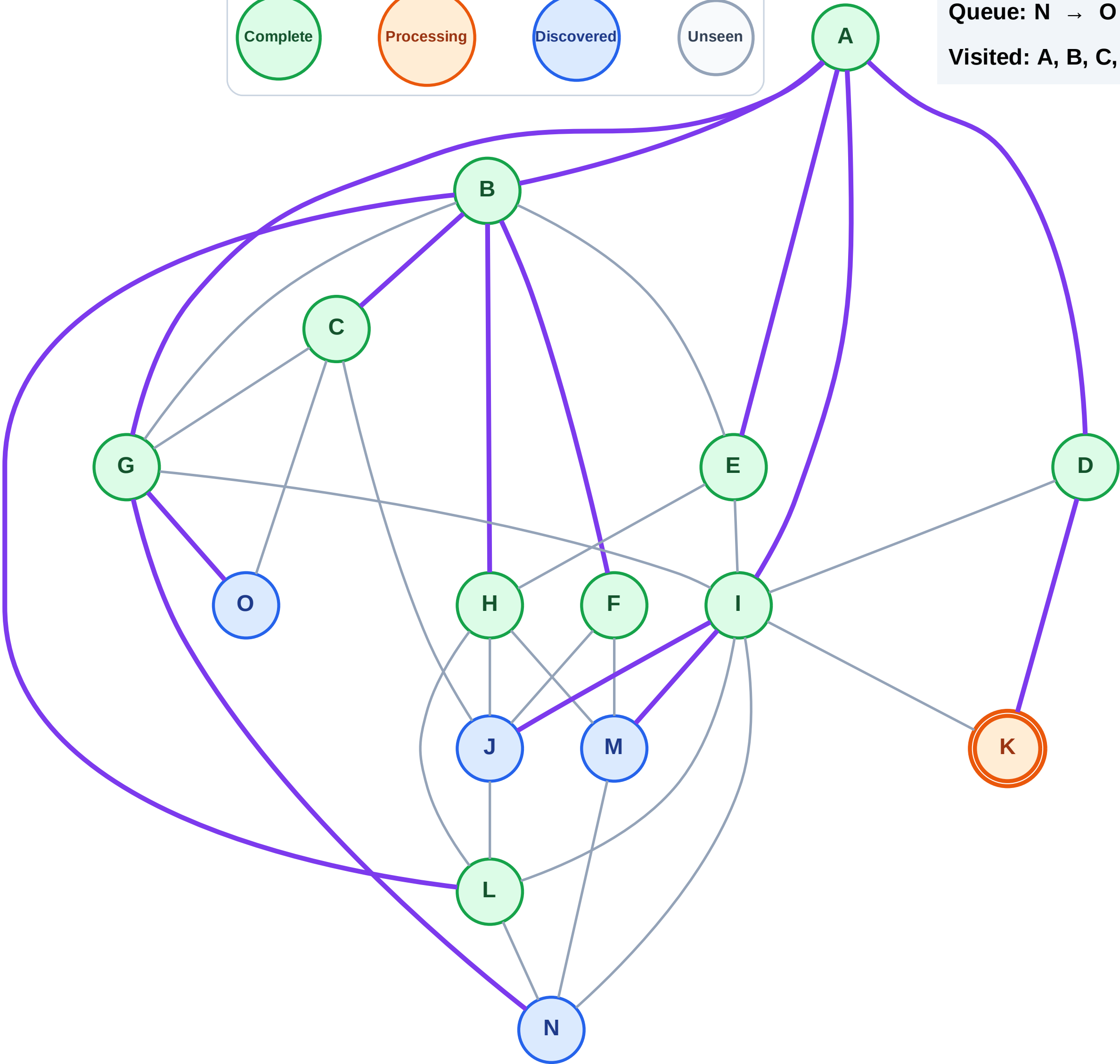
Processing

Discovered

Unseen

Queue: N → O → J → M

Visited: A, B, C, D, E, F, G, H, I, K, L



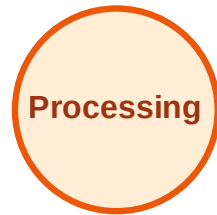
# DFS Traversal

Step 1: Start at A (Seeking target node K)

## Legend



Complete



Processing



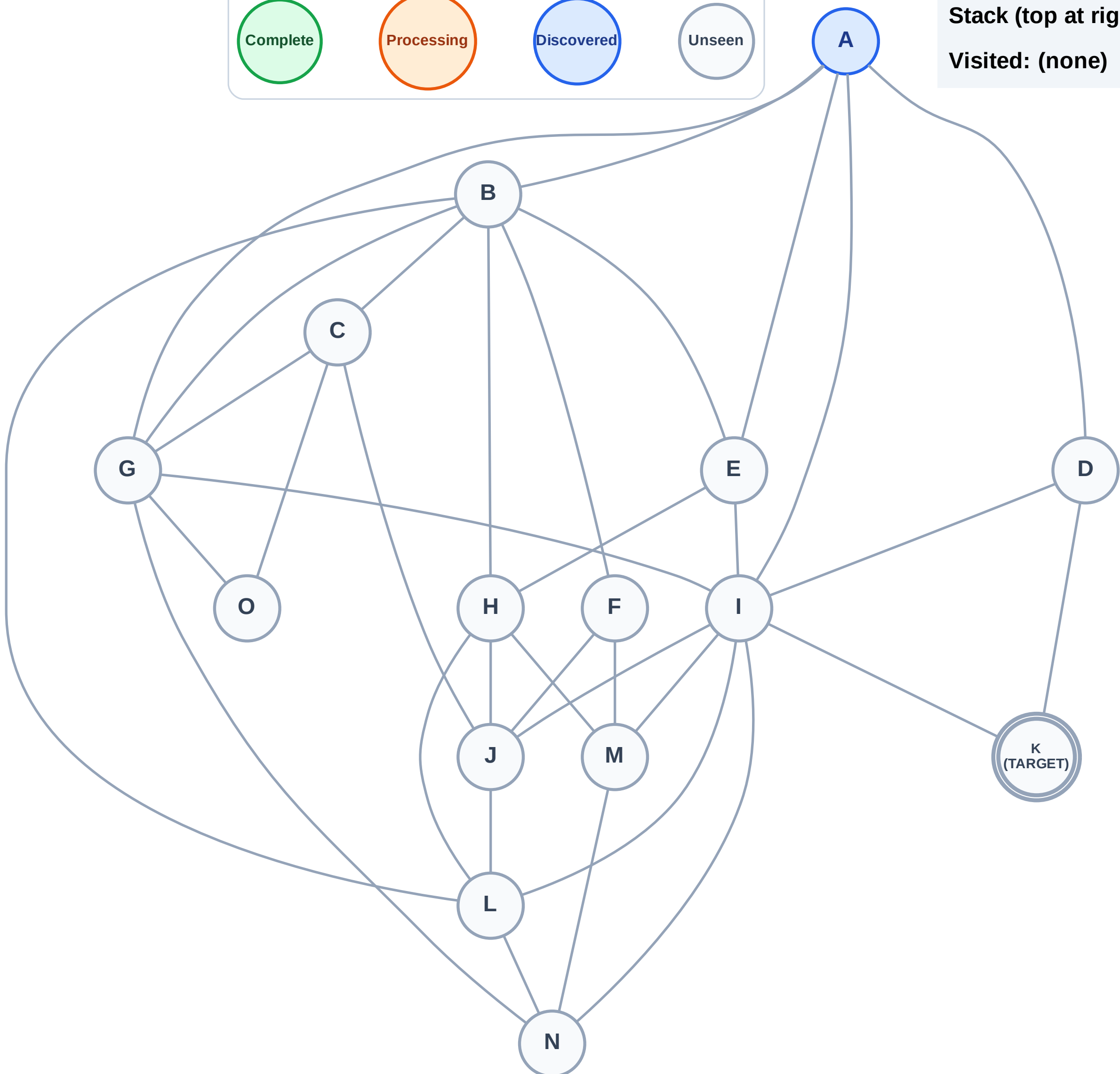
Discovered



Unseen

Stack (top at right): A

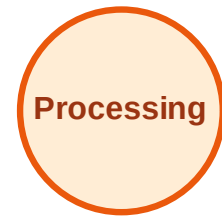
Visited: (none)



# DFS Traversal

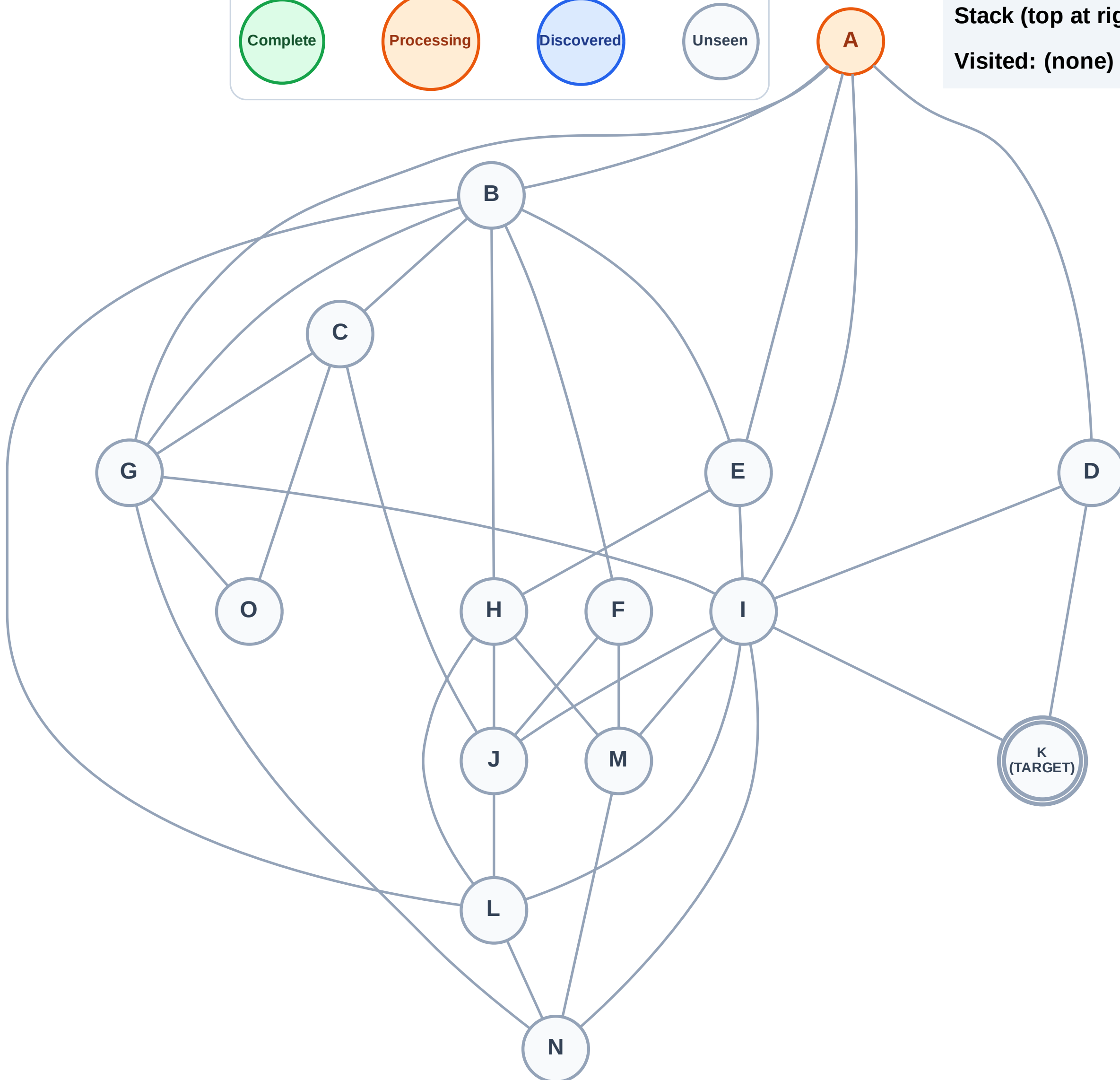
Step 2: Process node A

## Legend



Stack (top at right): (empty)

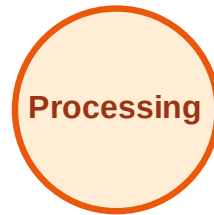
Visited: (none)



# DFS Traversal

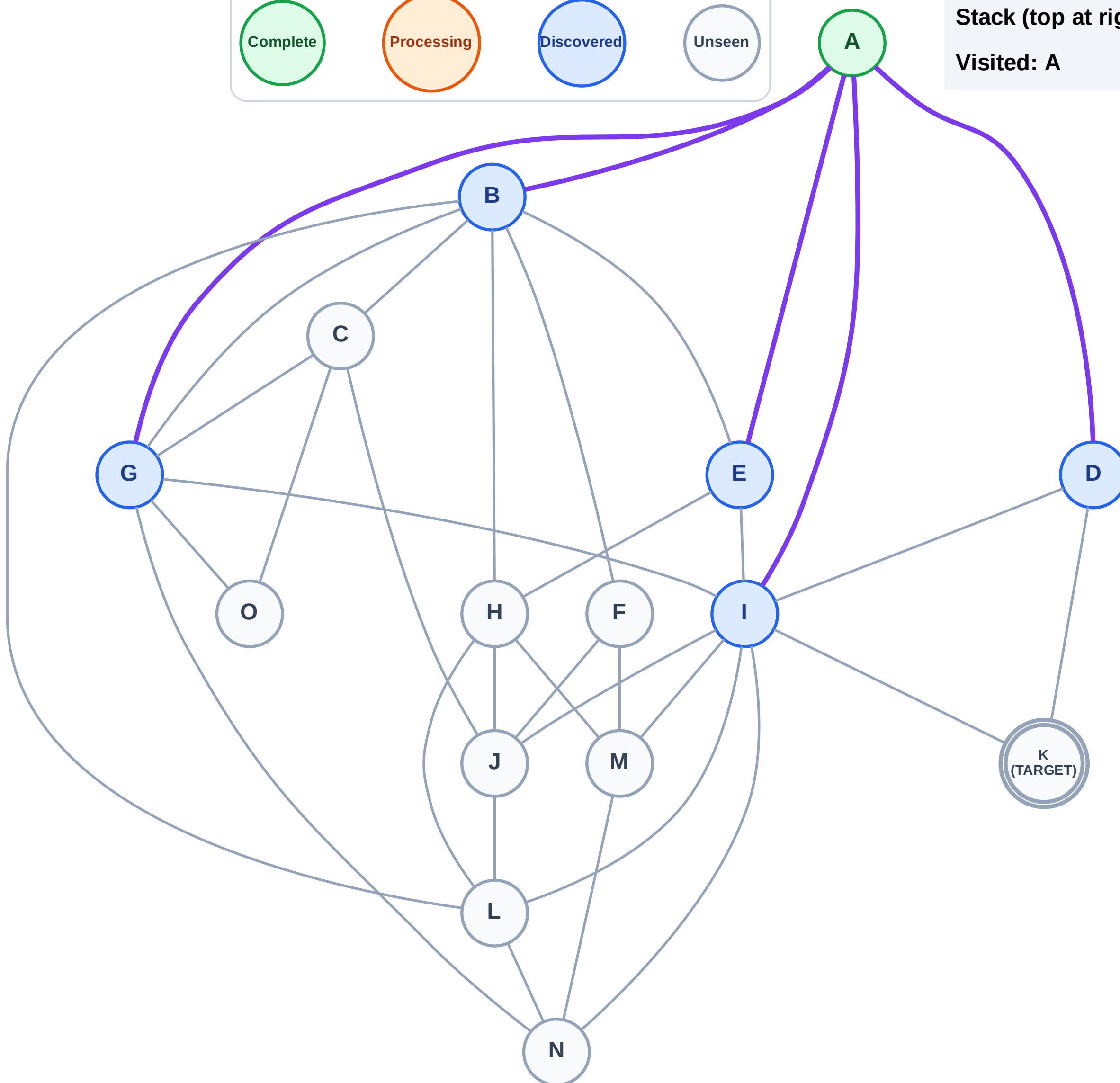
Step 3: Discover I, G, E, D, B from A

## Legend



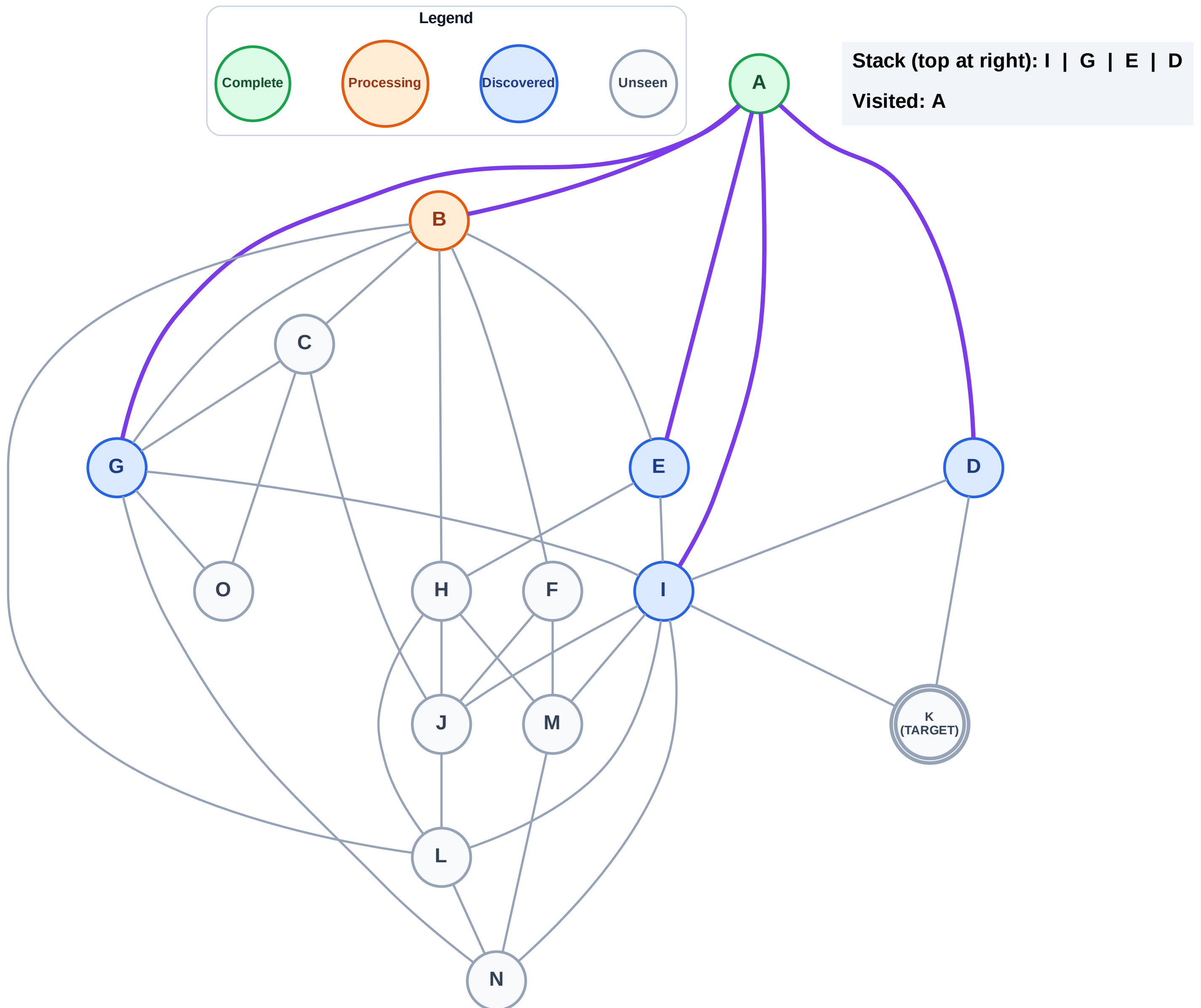
Stack (top at right): I | G | E | D | B

Visited: A



### Step 4: Process node B

### Step 4: Process node B



# DFS Traversal

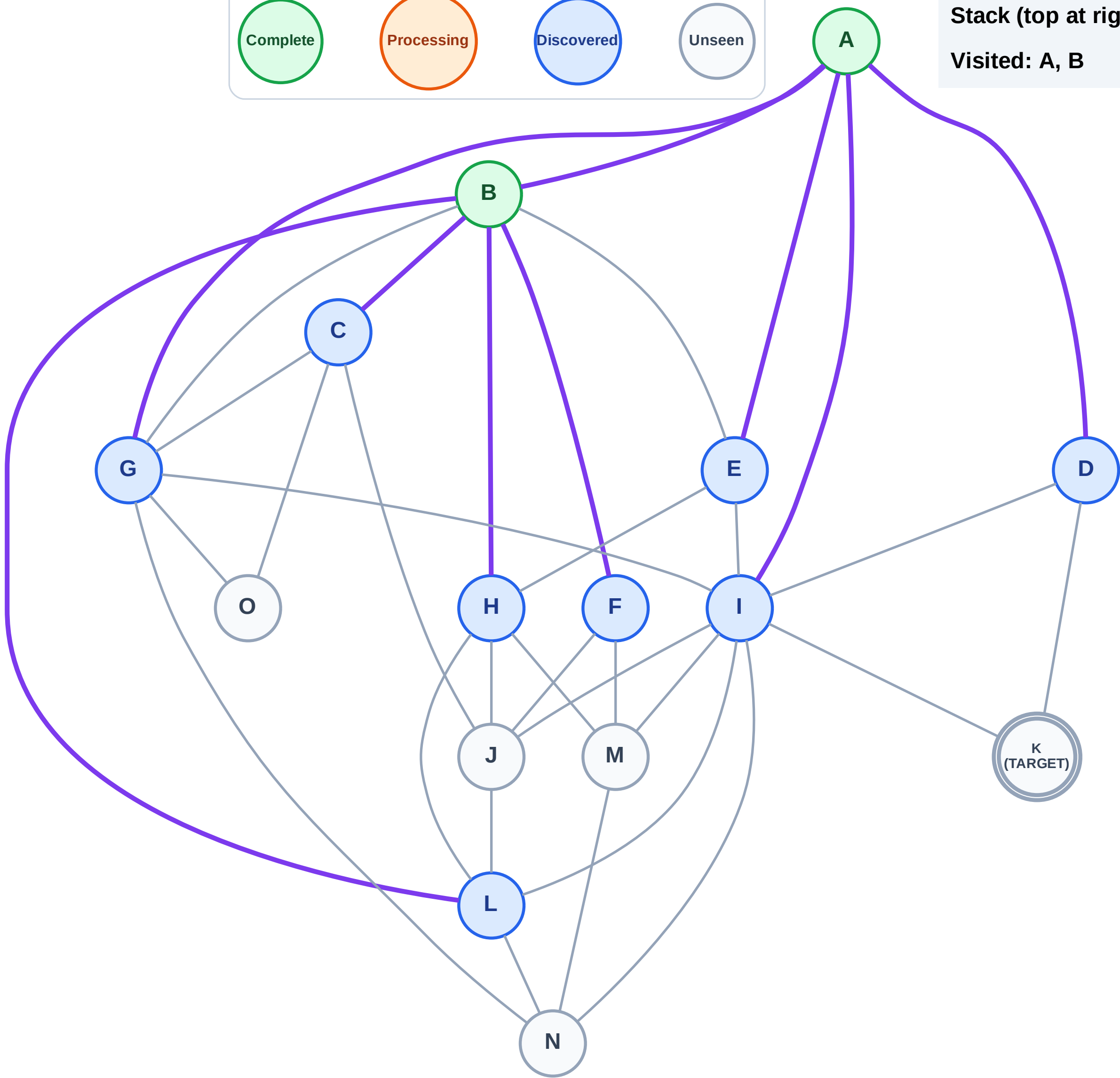
Step 5: Discover L, H, F, C from B

## Legend



Stack (top at right): I | G | E | D | L | H | F | C

Visited: A, B



# DFS Traversal

Step 6: Process node C

Legend

Complete

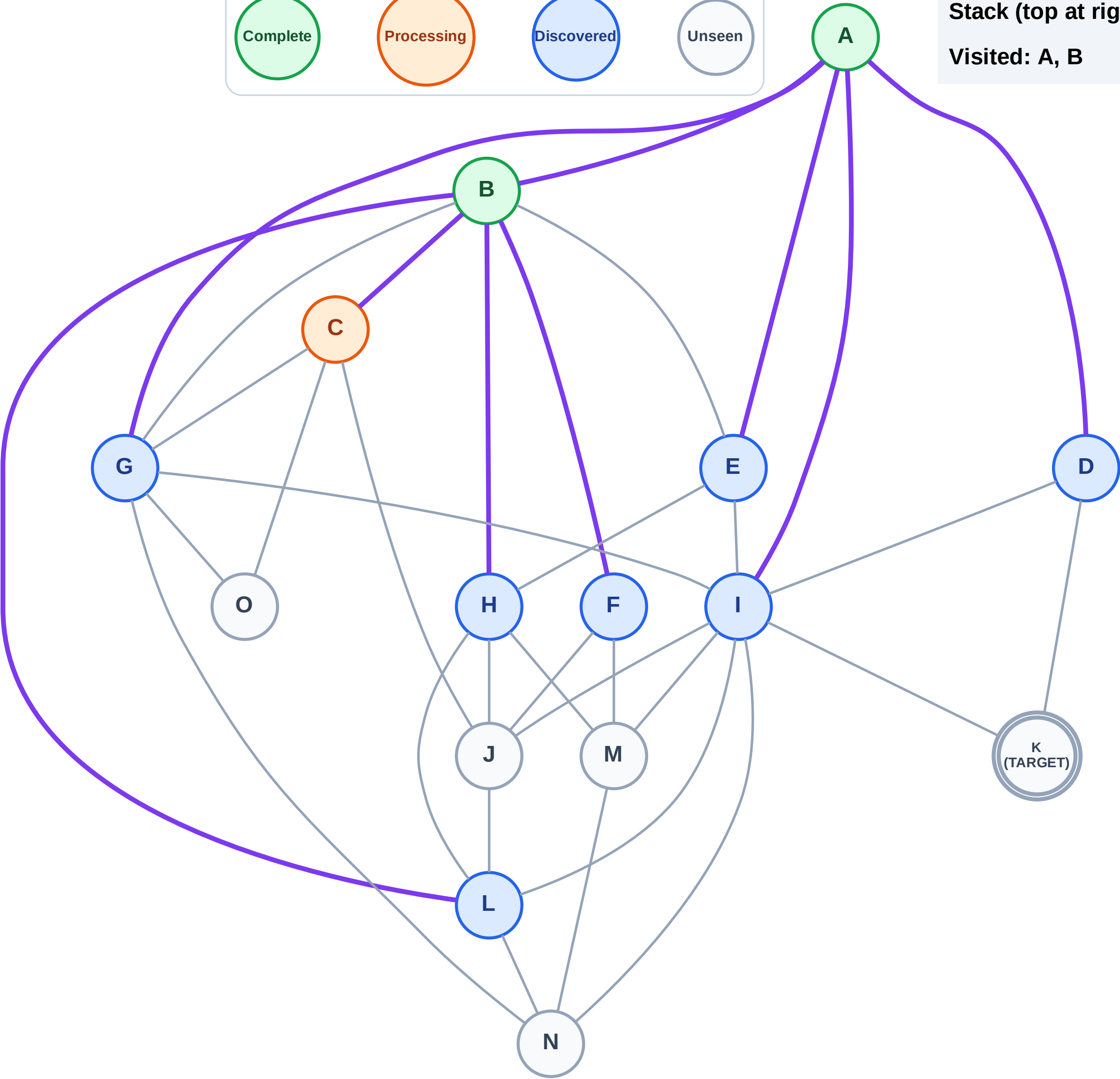
Processing

Discovered

Unseen

Stack (top at right): I | G | E | D | L | H | F

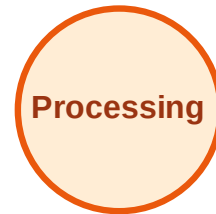
Visited: A, B



# DFS Traversal

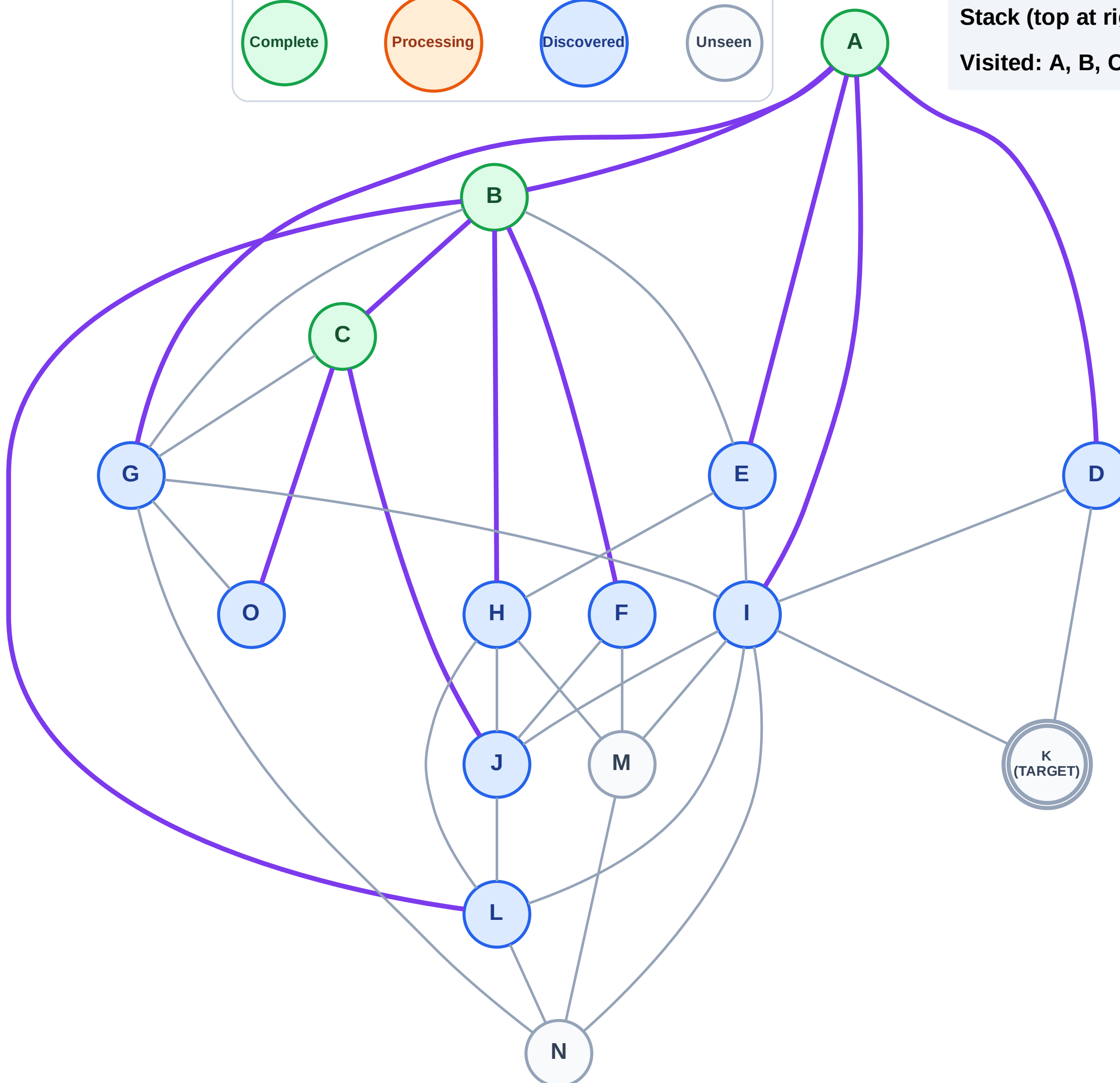
Step 7: Discover O, J from C

## Legend



Stack (top at right): I | G | E | D | L | H | F | O | J

Visited: A, B, C



# DFS Traversal

Step 8: Process node J

Legend

Complete

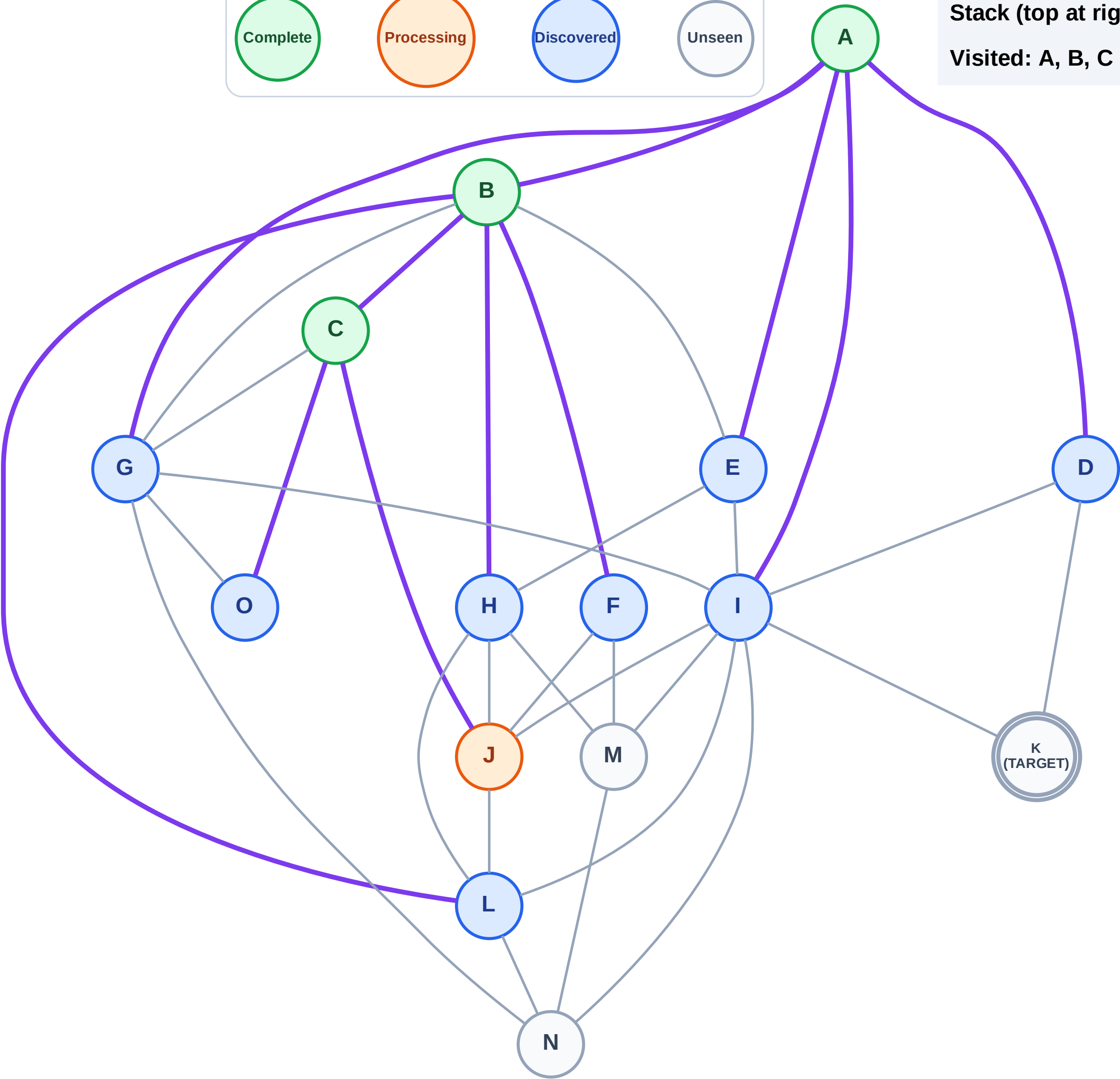
Processing

Discovered

Unseen

Stack (top at right): I | G | E | D | L | H | F | O

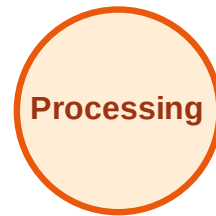
Visited: A, B, C



# DFS Traversal

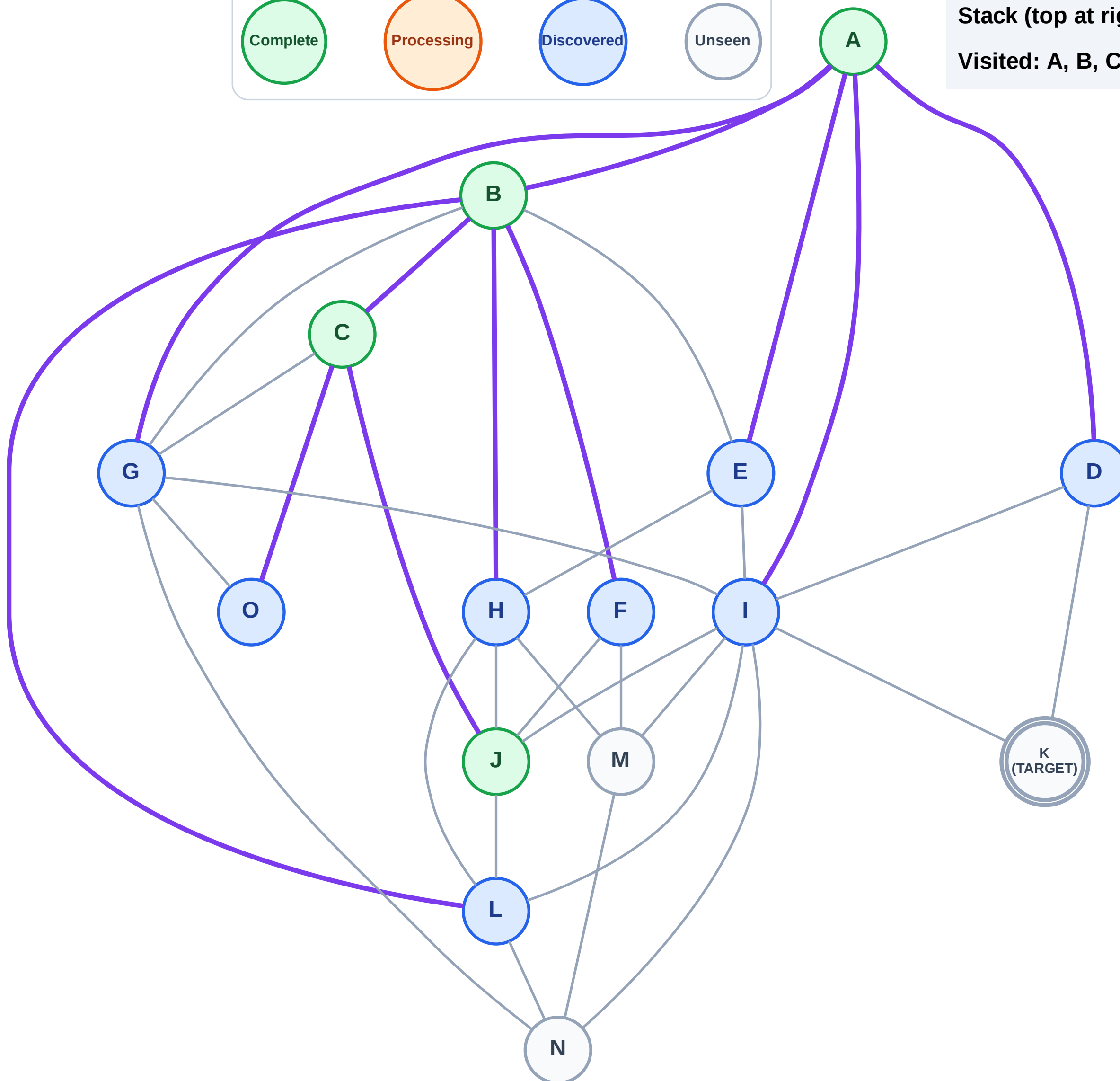
Step 9: No new neighbors from J

## Legend



Stack (top at right): I | G | E | D | L | H | F | O

Visited: A, B, C, J



# DFS Traversal

Step 10: Process node O

## Legend



Complete



Processing



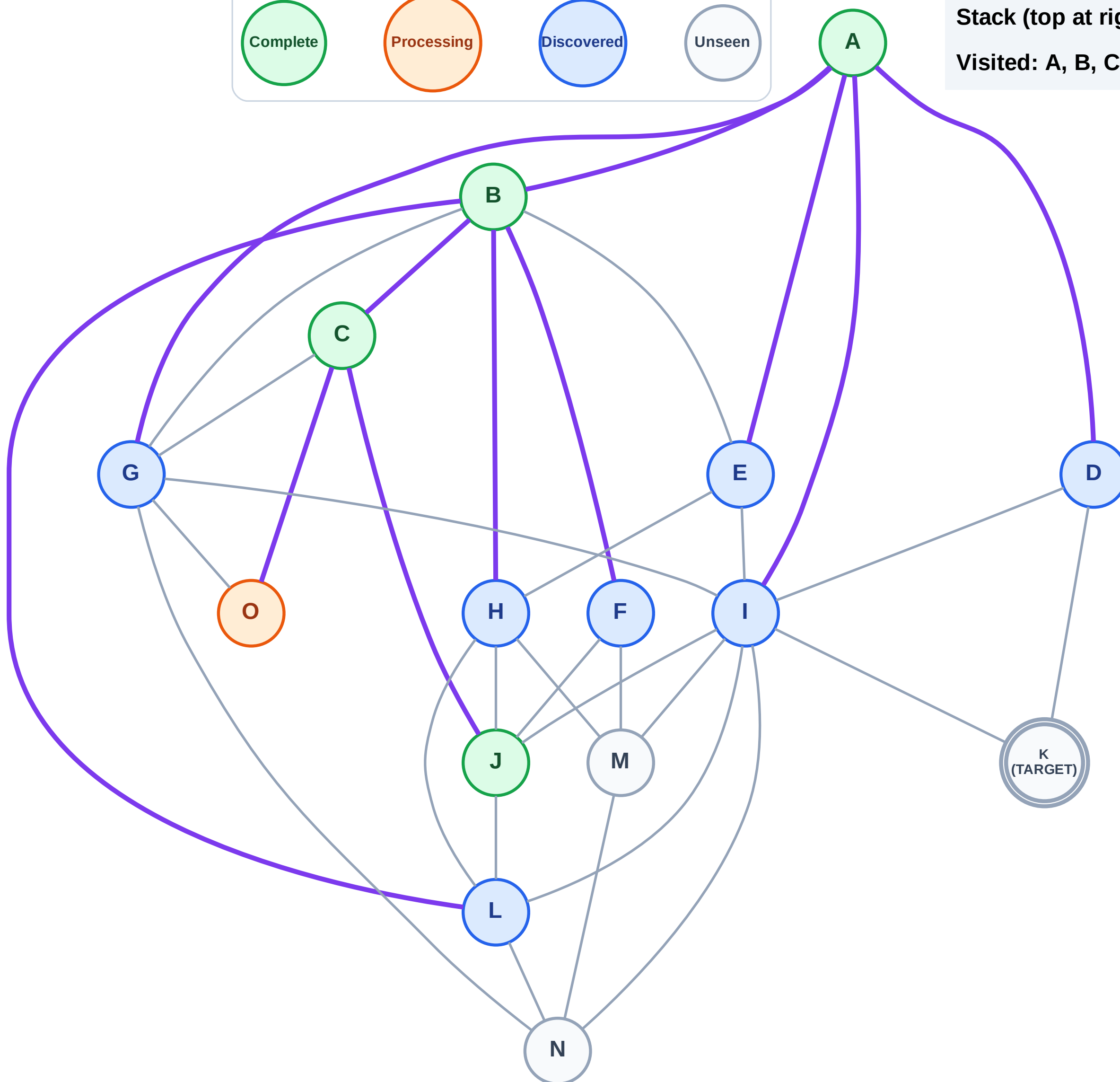
Discovered



Unseen

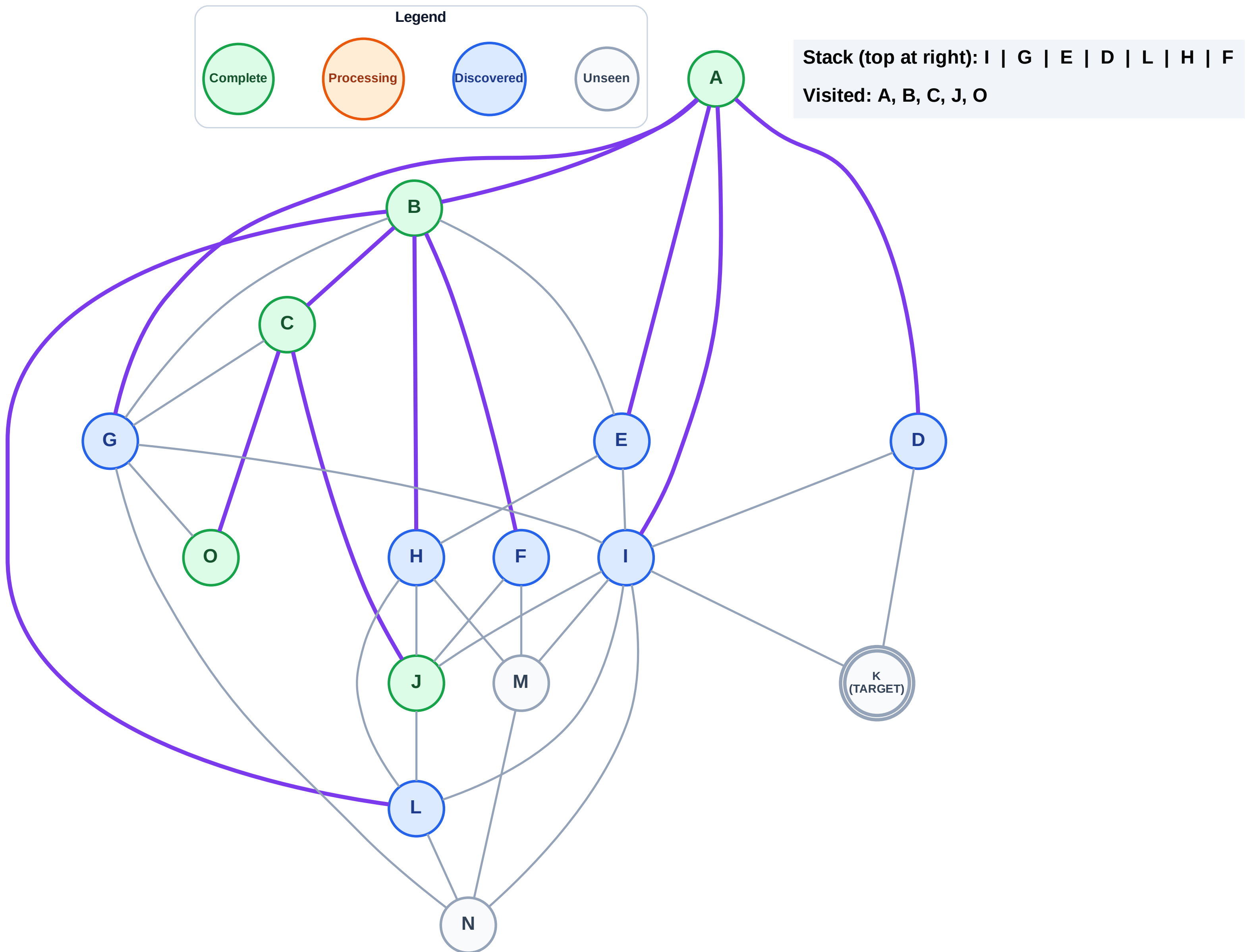
Stack (top at right): I | G | E | D | L | H | F

Visited: A, B, C, J



## Step 11: No new neighbors from O

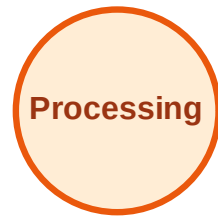
## Step 11: No new neighbors from O



# DFS Traversal

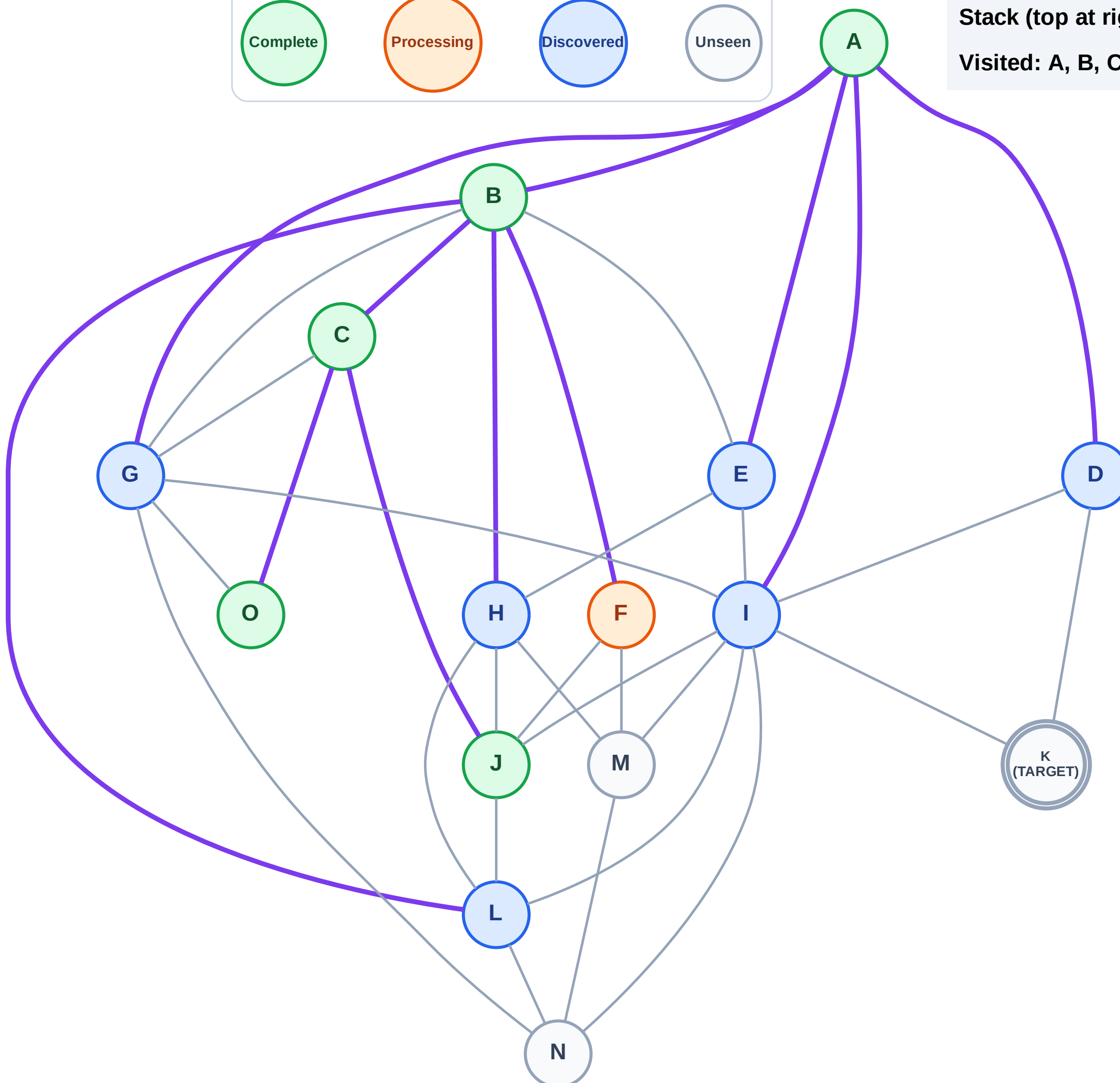
Step 12: Process node F

## Legend



Stack (top at right): I | G | E | D | L | H

Visited: A, B, C, J, O



# DFS Traversal

Step 13: Discover M from F

## Legend



Complete



Processing



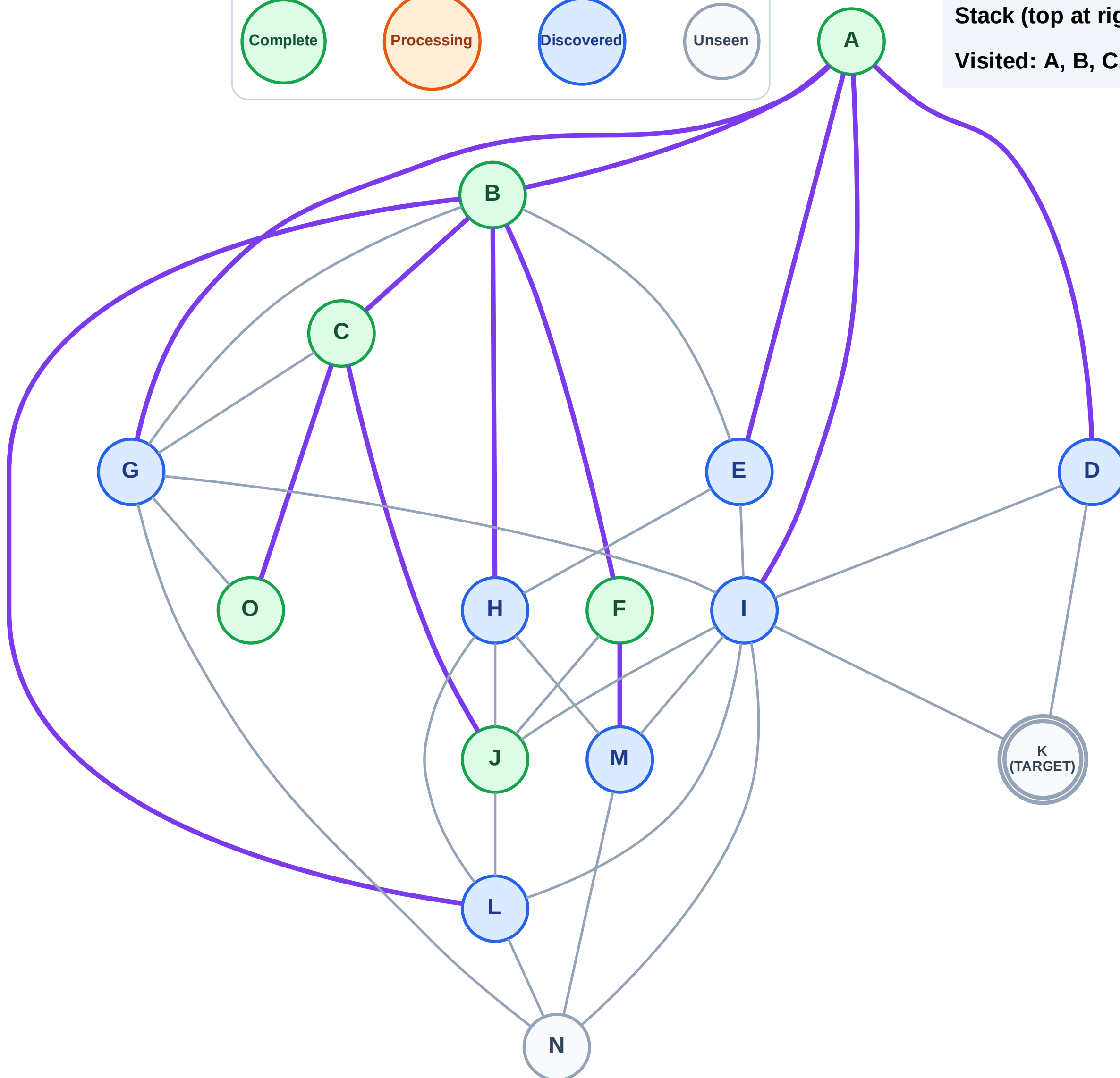
Discovered



Unseen

Stack (top at right): I | G | E | D | L | H | M

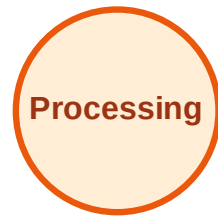
Visited: A, B, C, F, J, O



# DFS Traversal

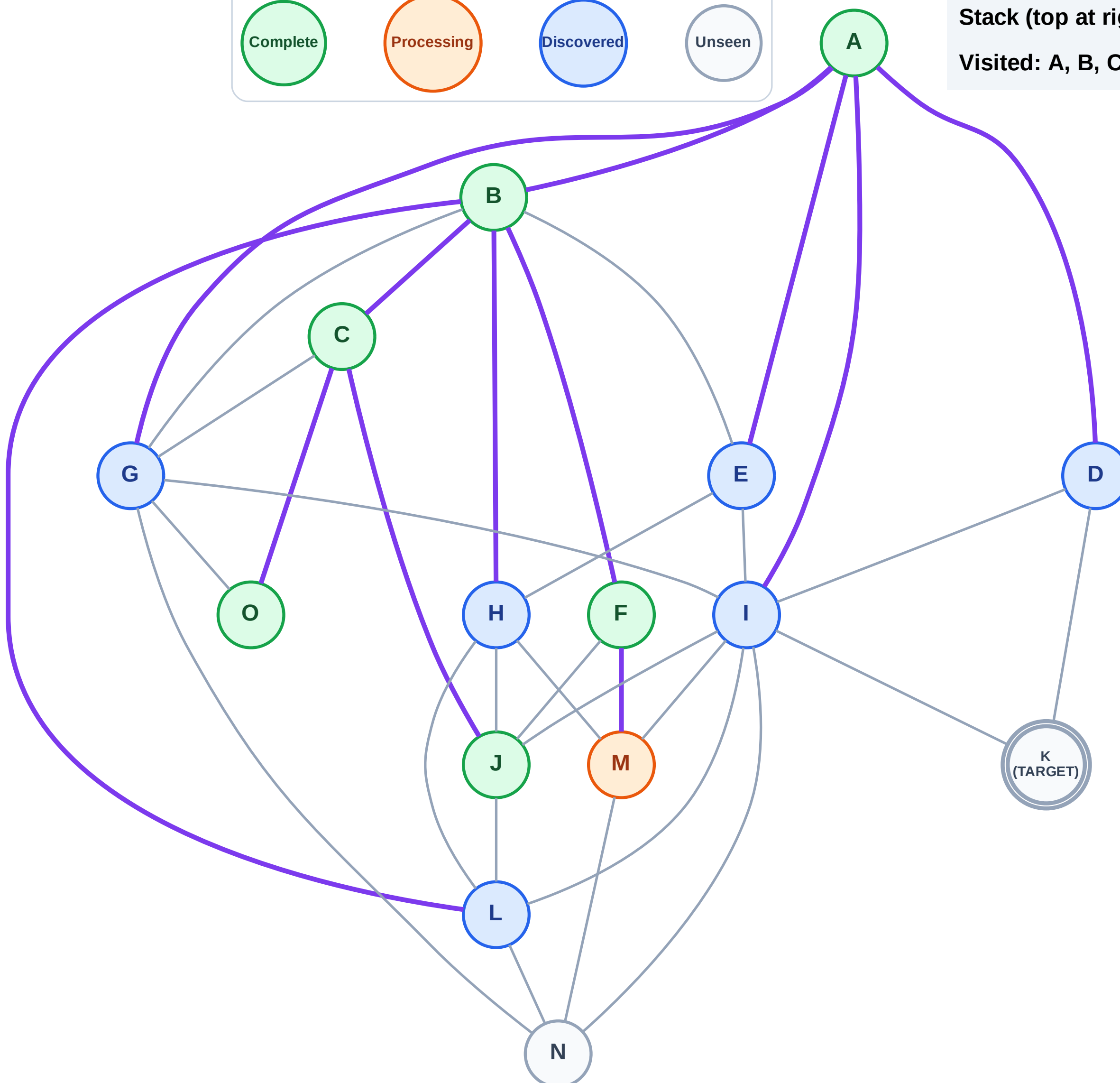
Step 14: Process node M

## Legend



Stack (top at right): I | G | E | D | L | H

Visited: A, B, C, F, J, O



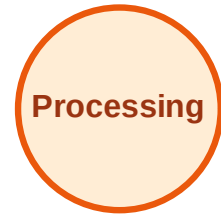
# DFS Traversal

Step 15: Discover N from M

## Legend



Complete



Processing



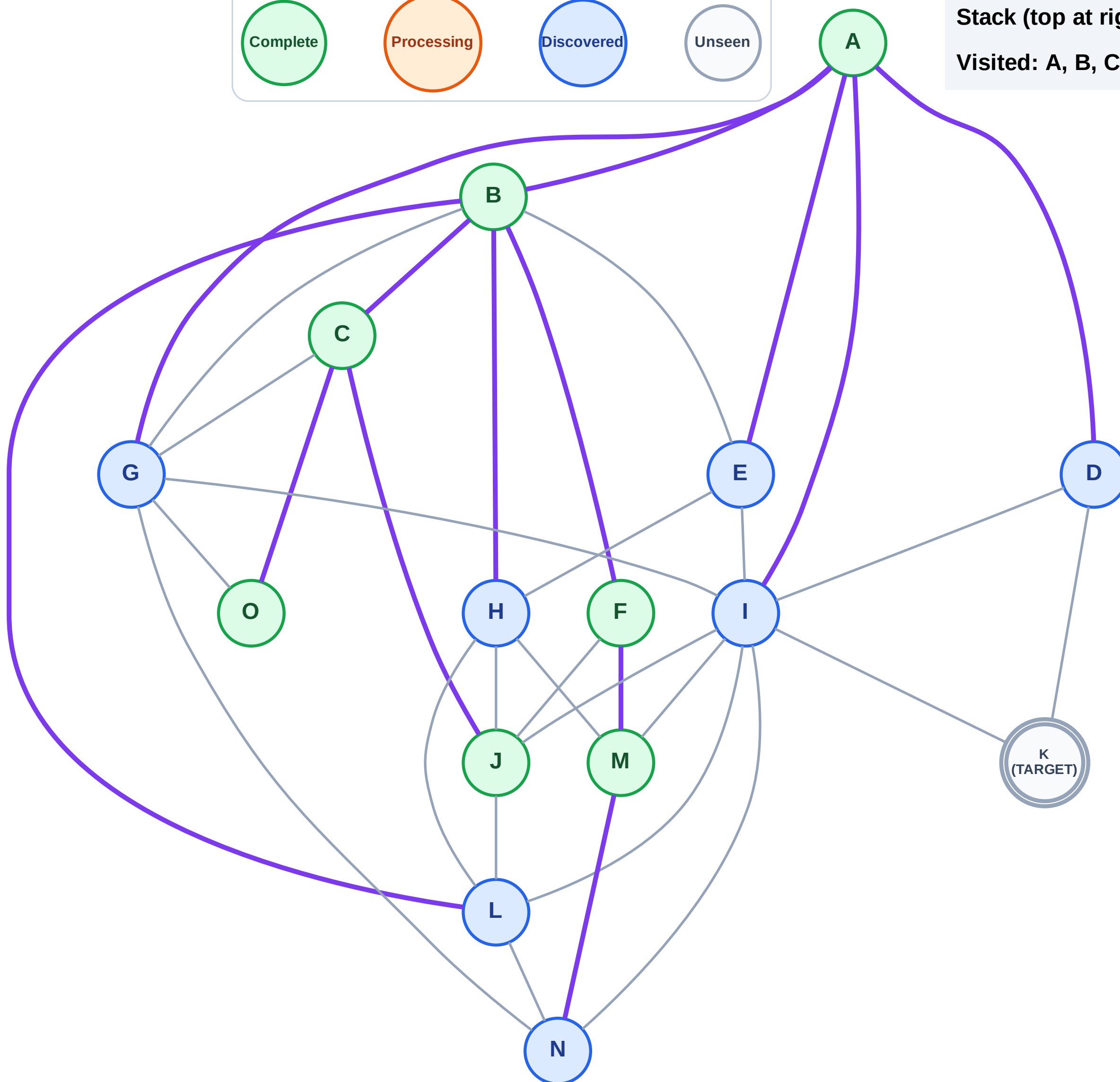
Discovered



Unseen

Stack (top at right): I | G | E | D | L | H | N

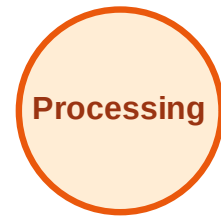
Visited: A, B, C, F, J, M, O



# DFS Traversal

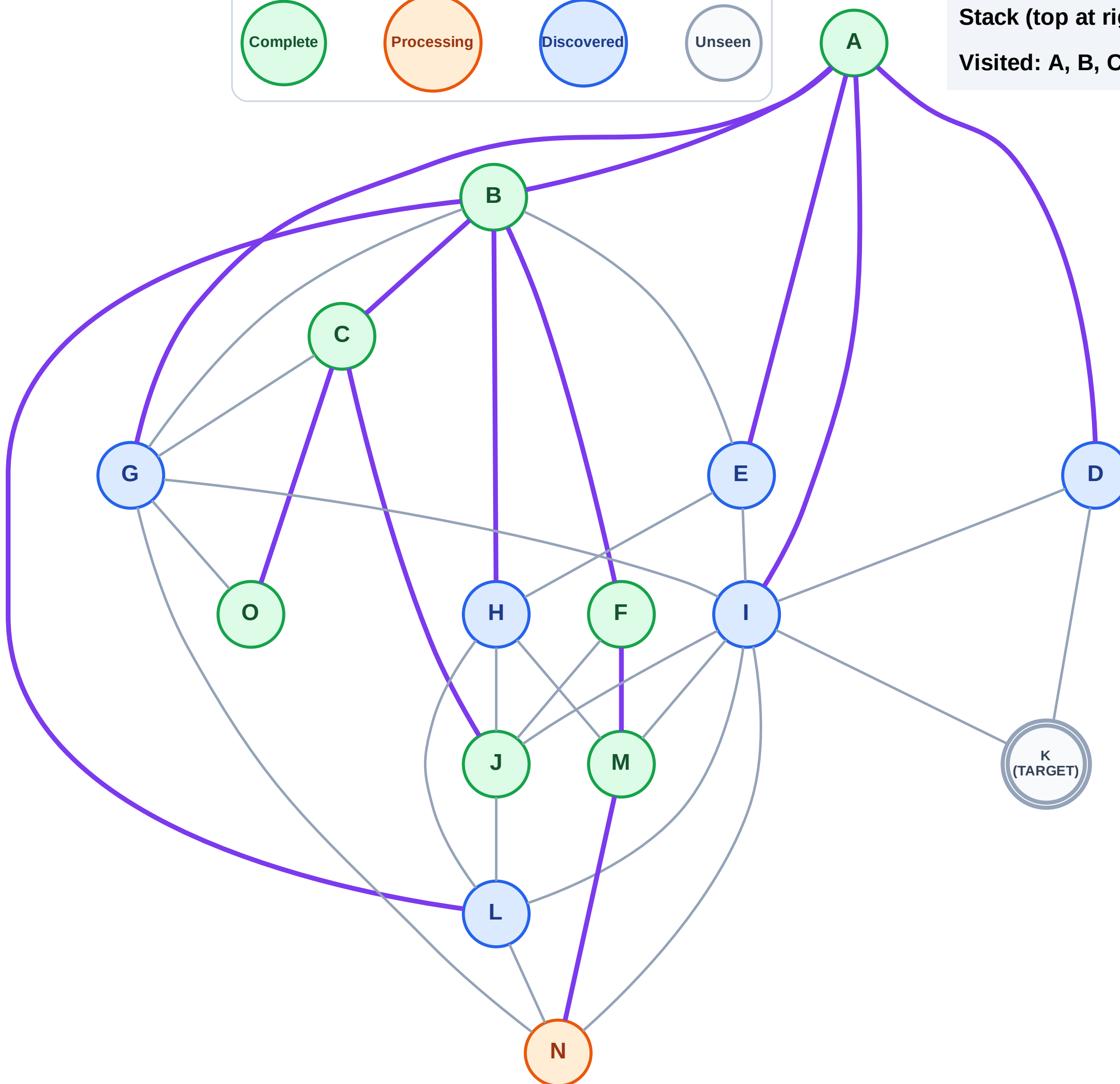
Step 16: Process node N

## Legend



Stack (top at right): I | G | E | D | L | H

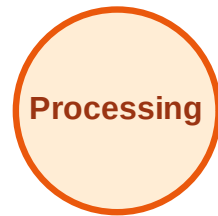
Visited: A, B, C, F, J, M, O



# DFS Traversal

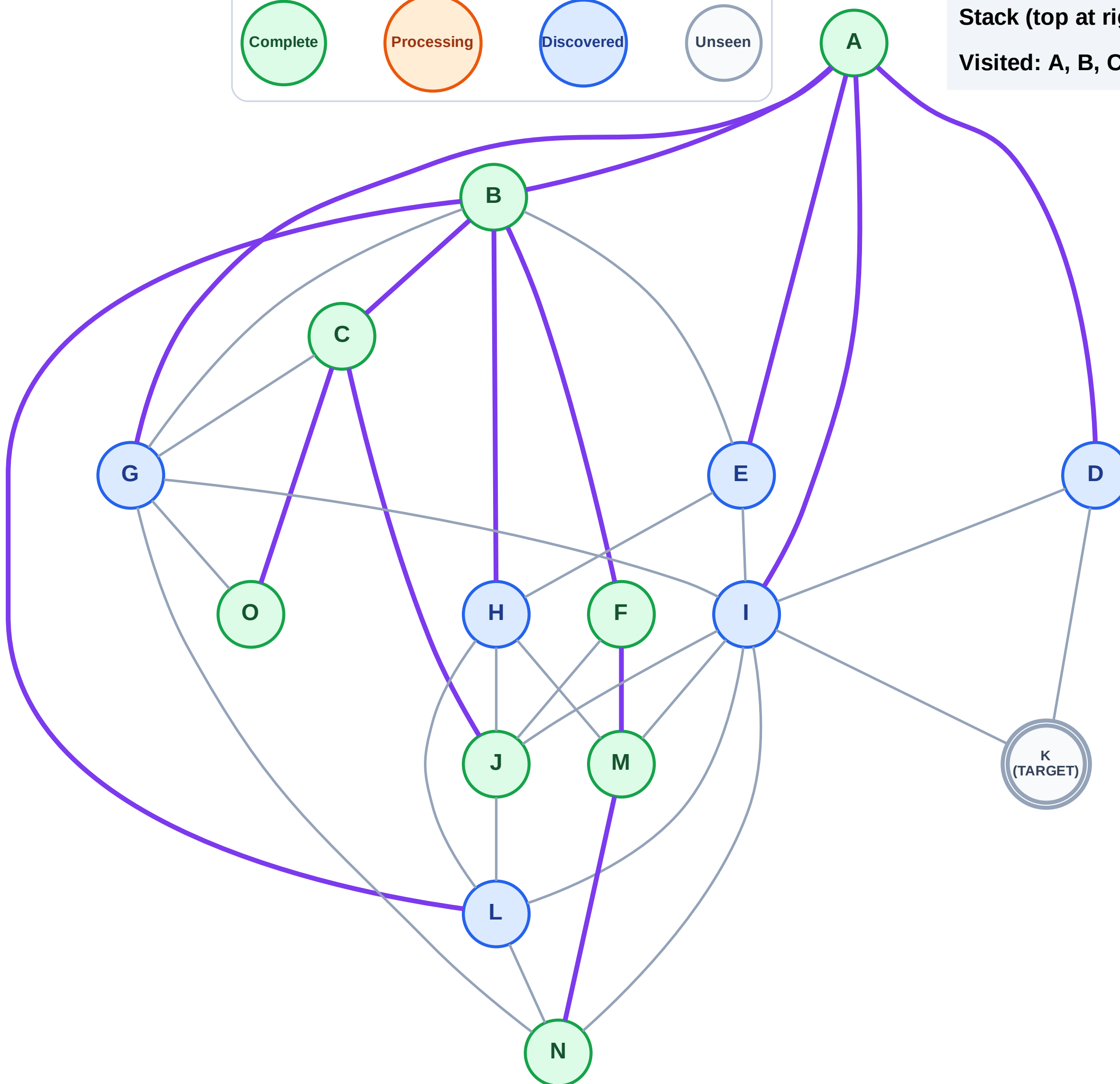
Step 17: No new neighbors from N

## Legend



Stack (top at right): I | G | E | D | L | H

Visited: A, B, C, F, J, M, N, O



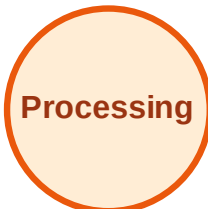
# DFS Traversal

Step 18: Process node H

## Legend



Complete



Processing



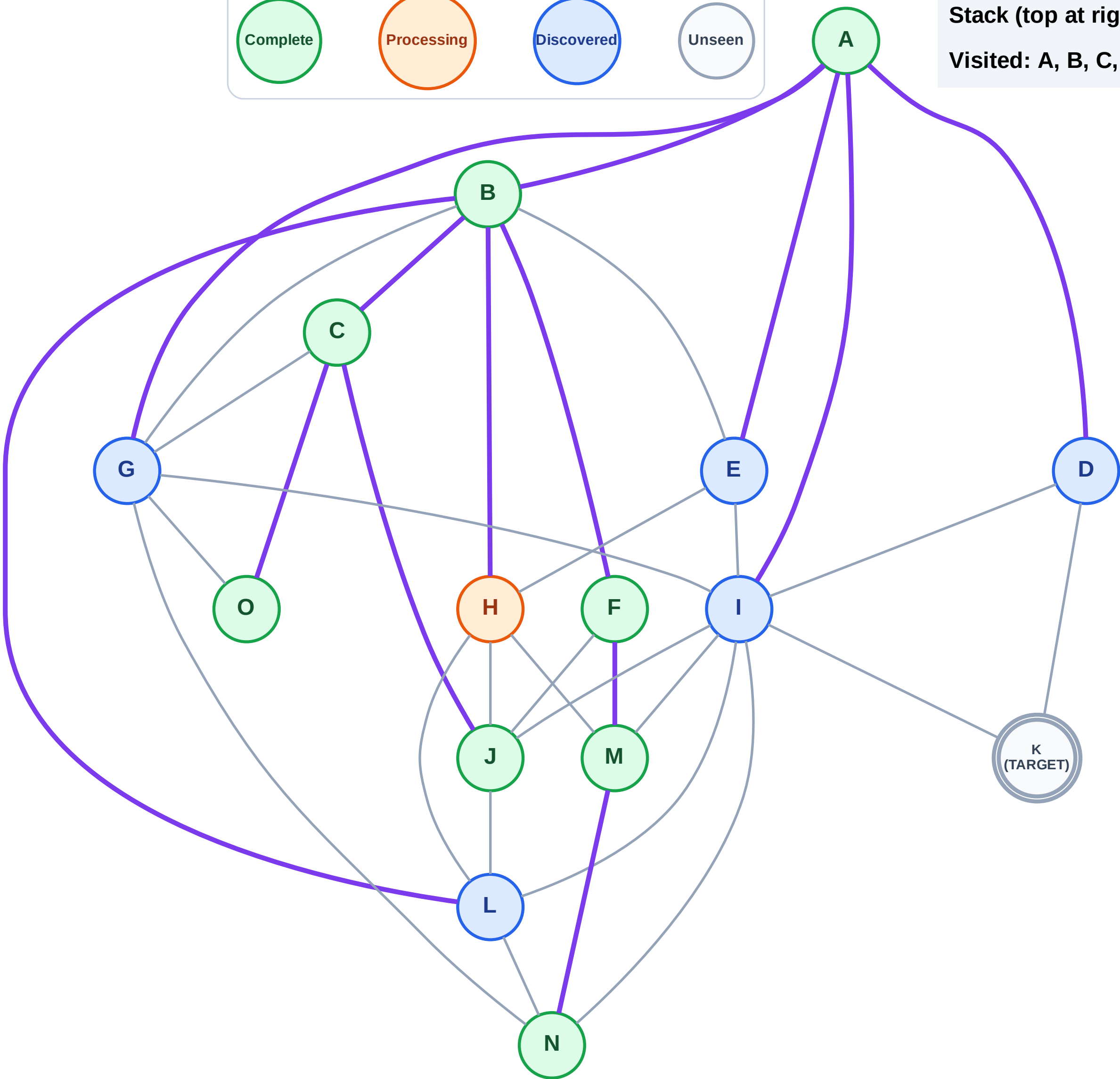
Discovered



Unseen

Stack (top at right): I | G | E | D | L

Visited: A, B, C, F, J, M, N, O



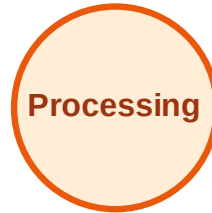
# DFS Traversal

Step 19: No new neighbors from H

## Legend



Complete



Processing



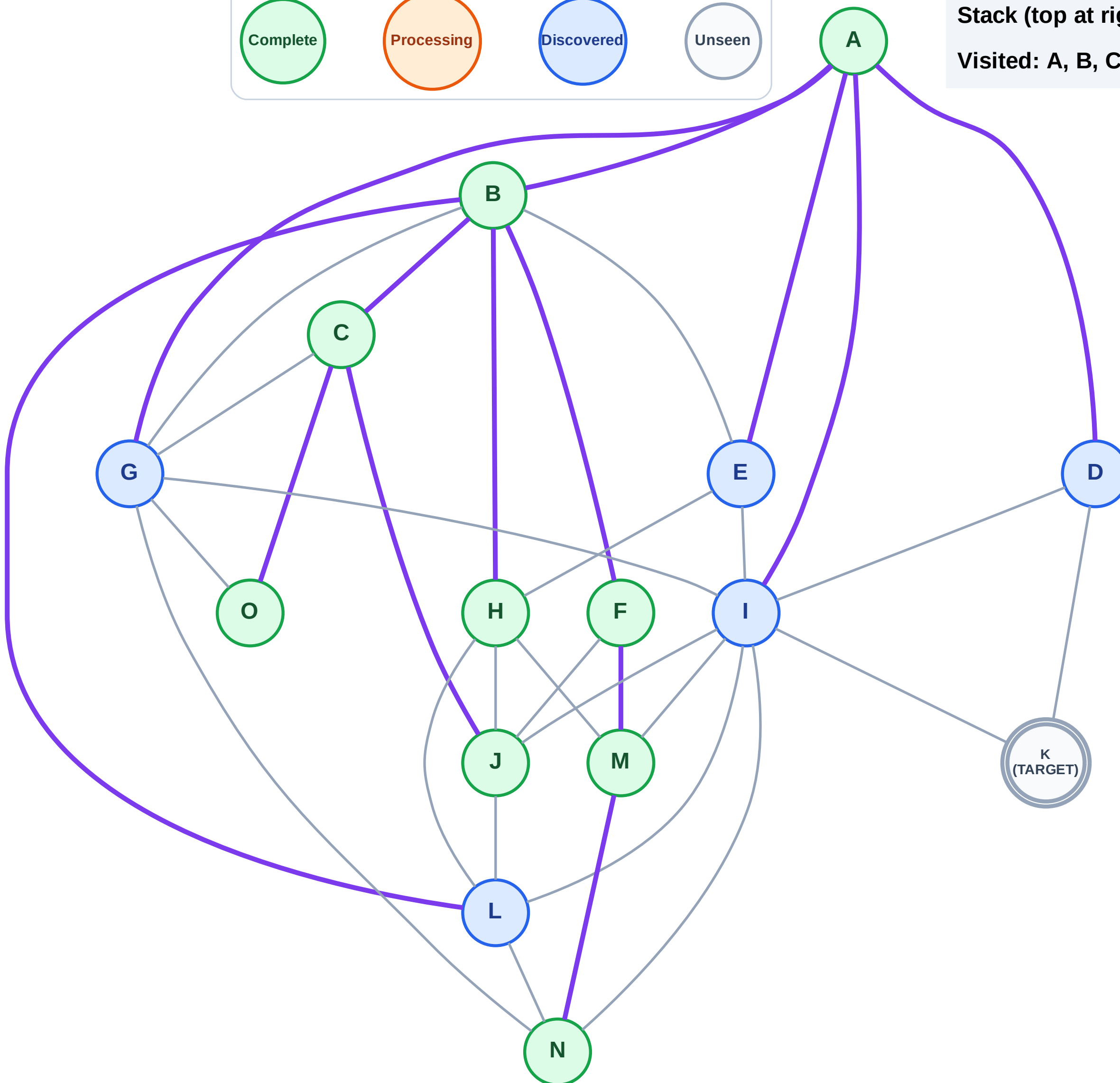
Discovered



Unseen

Stack (top at right): I | G | E | D | L

Visited: A, B, C, F, H, J, M, N, O



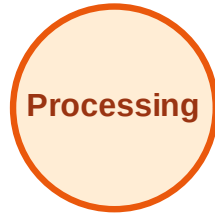
# DFS Traversal

Step 20: Process node L

## Legend



Complete



Processing



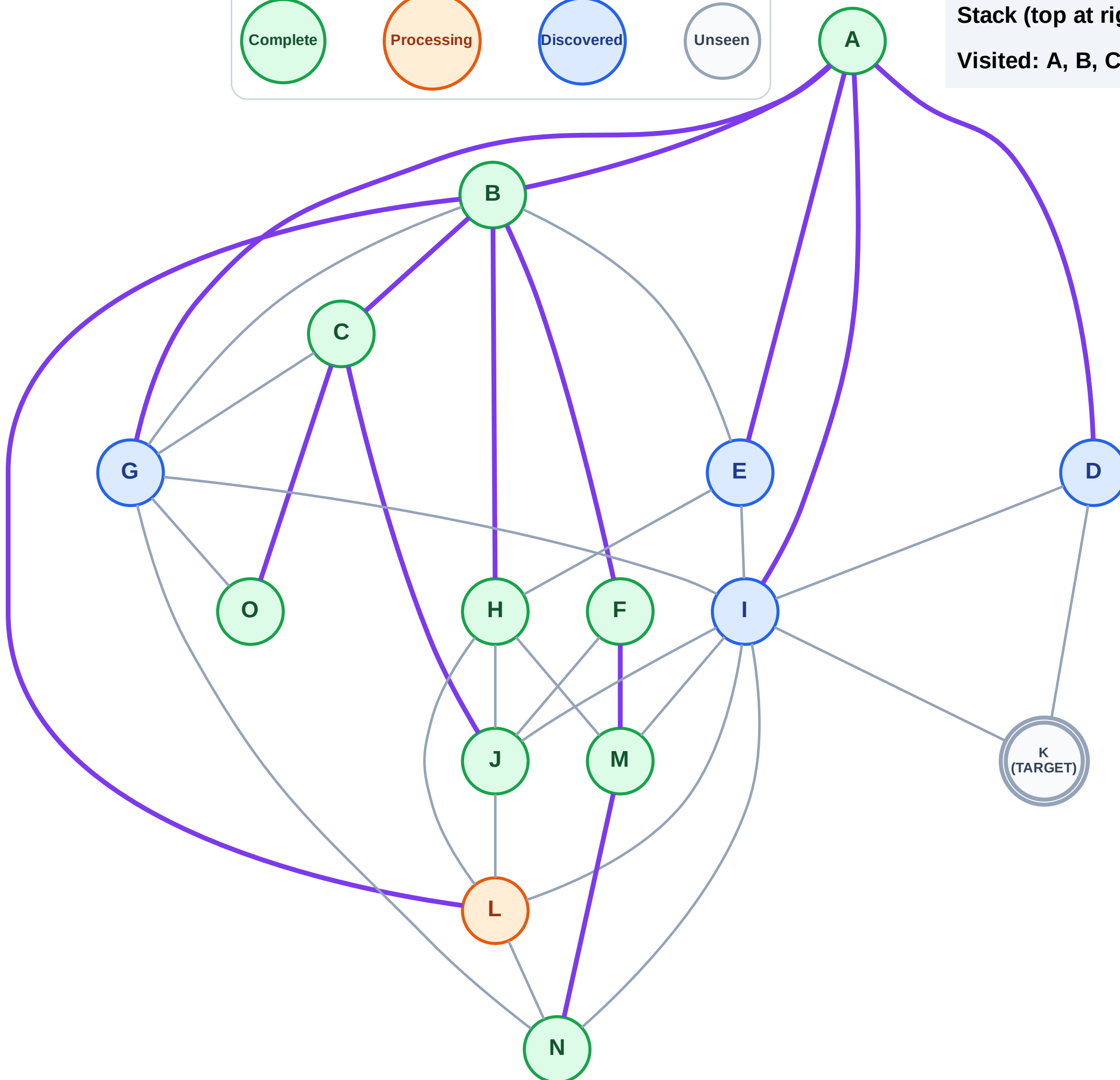
Discovered



Unseen

Stack (top at right): I | G | E | D

Visited: A, B, C, F, H, J, M, N, O



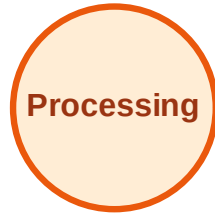
# DFS Traversal

Step 21: No new neighbors from L

## Legend



Complete



Processing



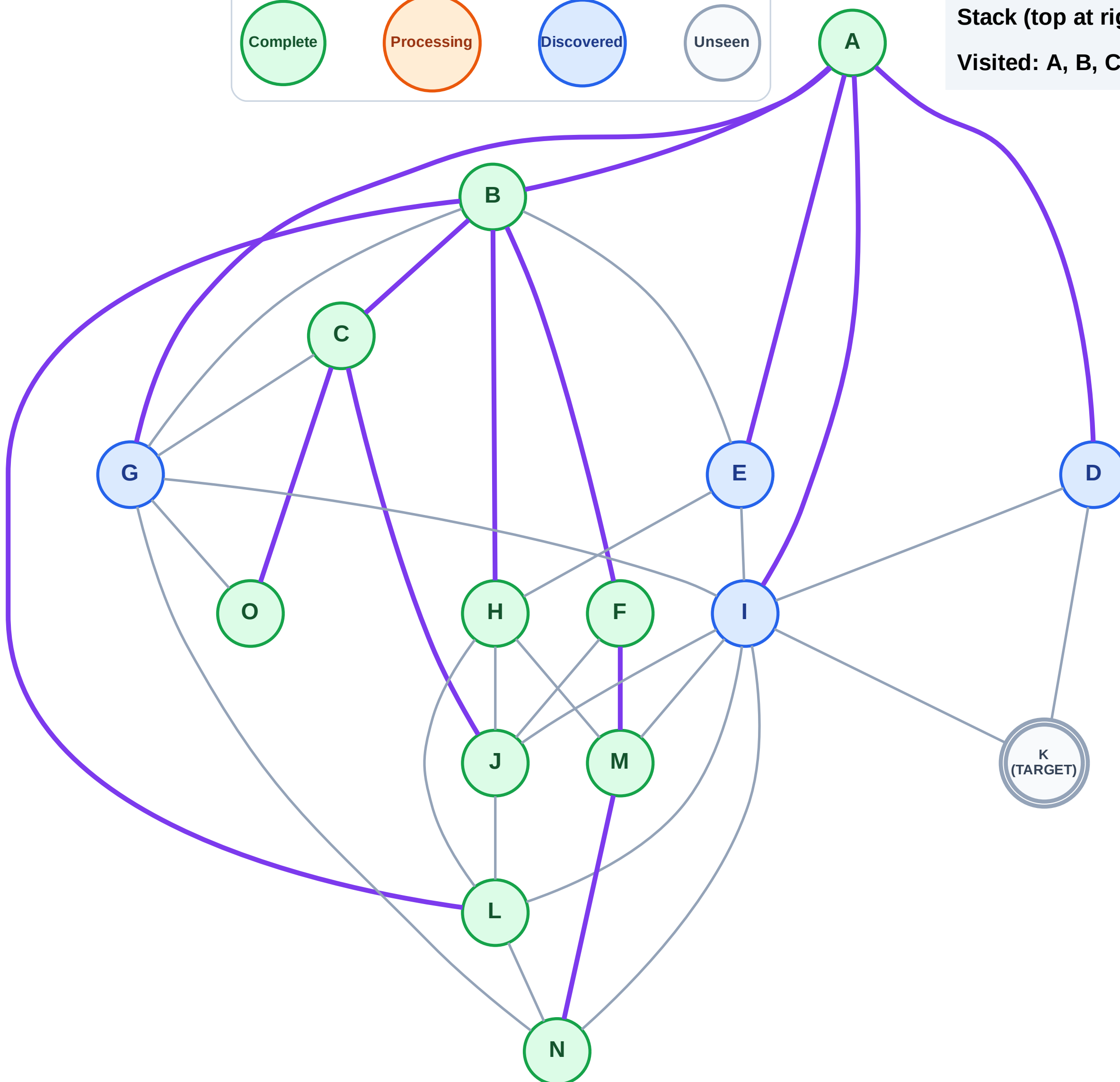
Discovered



Unseen

Stack (top at right): I | G | E | D

Visited: A, B, C, F, H, J, L, M, N, O



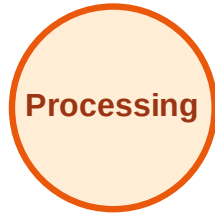
# DFS Traversal

Step 22: Process node D

## Legend



Complete



Processing

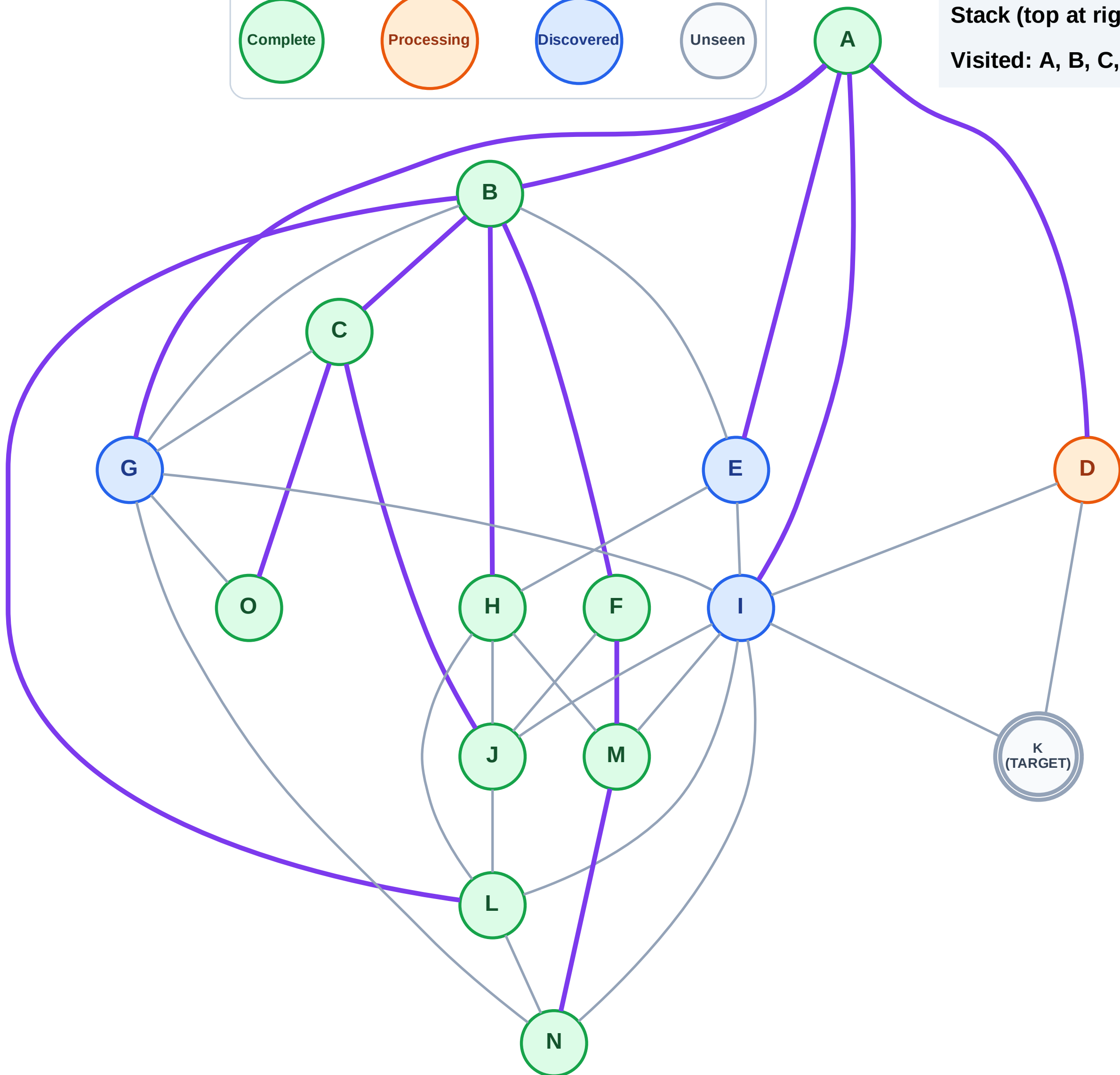


Discovered



Unseen

Stack (top at right): I | G | E  
Visited: A, B, C, F, H, J, L, M, N, O



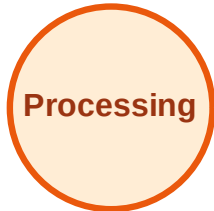
# DFS Traversal

Step 23: Discover K from D

## Legend



Complete



Processing



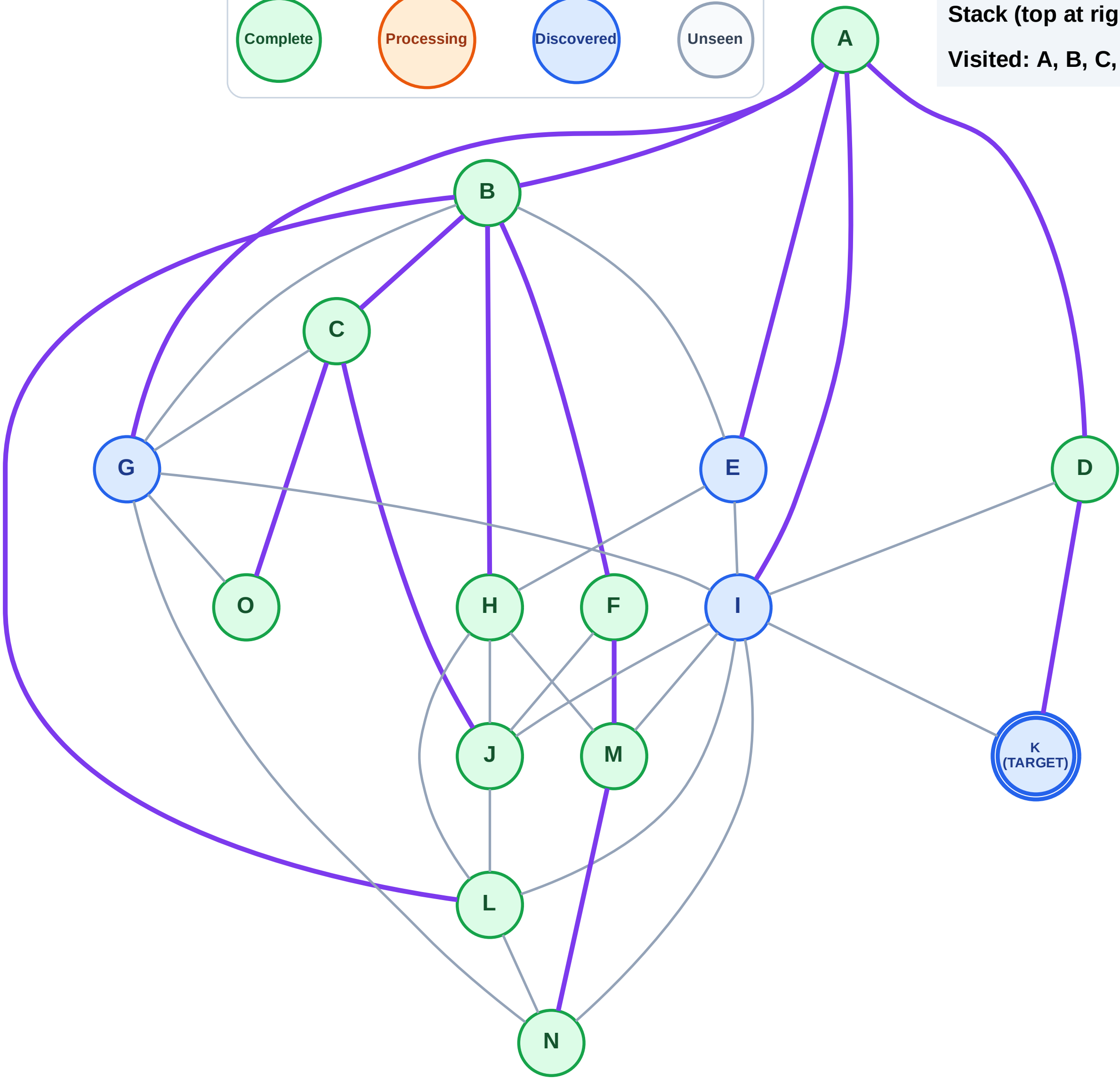
Discovered



Unseen

Stack (top at right): I | G | E | K

Visited: A, B, C, D, F, H, J, L, M, N, O

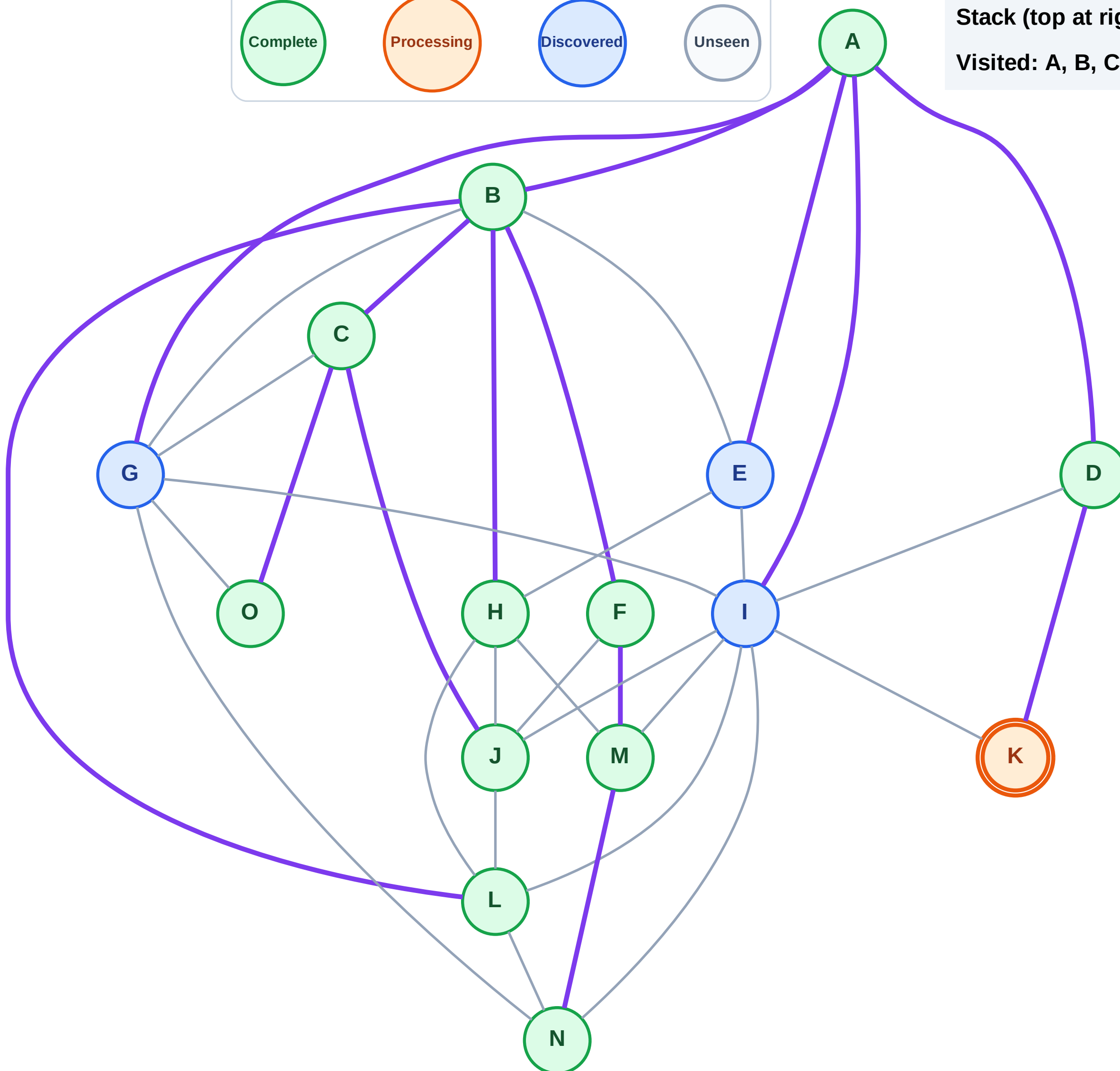


### Step 24: Process node K

### Legend

## Unseen

**Visited: A, B, C, D, F, H, J, L, M, N, O**



# DFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): I | G | E  
Visited: A, B, C, D, F, H, J, K, L, M, N, O

